BERNSTEIN
SOCIETE GENERALE GROUP

21 July 2026

## Global Energy Storage

# Global Energy Storage: Data center battery intensity rises as sentiment cools


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**Data centers are becoming more battery intense with average duration increasing to c. 5hrs as more are built behind the meter**. Batteries are used in data centers for centralized power (UPS), distributed power (BBU) and energy storage (ESS). While UPS and BBU demand is relatively short duration (<0.25hrs), ESS demand can be longer. With 24% of data centers under construction off grid and 39% of the pipeline off grid, battery energy storage is rising in duration with some hyperscalers having up to 8hours storage.

**Assuming 103GW of data center capacity by 2030, cumulative battery capacity will rise to 277GWh.** We assume US data center capacity will increase from 41GW to 103GW by 2030. We expect annual data center battery demand to increase 10x from 5GWh to 66GWh. Cumulative ESS demand for data centers reached 11GWh in 2025 and will likely increase to 277GWh by 2030. This implies the battery /data center capacity GWh/GW multiplier will rise from 0.3 to 2.7x. Newer projects have a multiplier of 4-8x, implying further scope for battery demand to increase.

**Increased data center demand raises our total battery demand for the US to 486GWh by 2030 which represents 22% CAGR and a 15% increase on previous estimates.** 70% of incremental demand is forecast to be from ESS which will likely grow from 49GWh (2025) to 266GWh by 2030. ESS capacity in the US is currently limited at 36GWh but set to grow to 102GWh this year as more EV plants are converted to ESS.

**But while our base case has increased, it's also fair to say that so too has uncertainty in long term growth.** Increased data center capex and the success of lower cost Chinese models is creating near term uncertainty in the pace of AI data center build out. While the longer term outlook looks positive, there is a risk that capex could slow if hyperscalers become less confident in returns from the AI business model. Upcoming mid-year results from the hyperscalers may provide further guidance.

**Profitability remains another key question, with margins (ex-subsidy) close to breakeven.** While demand looks set to grow by 20-25% CAGR though to 2030, margins excluding tax benefits remain negative or close to zero. While guidance is for Korean battery maker margins to turn positive in 2H26, this remains still the subject of debate. While we think that higher ESS utilization will raise margins, continued de-rating of EV demand in the US YTD (-25%) may continue to put pressure on margins in the near term.

**Korean battery makers valuations have a better risk-reward (12x EV/EBITDA 2027 and 5x for 2030) but we would wait for a better entry point.** LGES and SDI are pricing in 22-26% revenue CAGR to end of decade (in line with industry growth) and a return to high single digit margins. Longer term we assume mid to high single digit growth and margins, which give c. 10% &30% upside from current levels for LGES and SDI respectively. While these stocks look more attractive, we would wait for a better entry point.


First Published: 20 Jul 2026 20:30 UTC Completion Date: 20 Jul 2026 12:39 UTC

## BERNSTEIN TICKER TABLE

| Ticker | Rating | Cur | 16 Jul 2026 Closing Price | Price Target | TTM Rel. Perf. | Cur | Reported EPS 2025A | Reported EPS 2026E | Reported EPS 2027E | Reported P/E (x) 2025A | Reported P/E (x) 2026E | Reported P/E (x) 2027E |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 373220.KS (LGES) | M | KRW | 332,000 | **351,000** | (29.1)% | KRW | (4,580.94) | **(2,364.33)** | **10,225** | (72.5) | (140.4) | 32.5 |
| *OLD* | | | | 347,000 | | | | 1,811.04 | 9,452.97 | | | |
| 051910.KS (LG Chem) | M | KRW | 260,000 | **312,000** | (34.9)% | KRW | (13,258.70) | **2,693.22** | **28,743** | (19.6) | 96.5 | 9.0 |
| *OLD* | | | | 298,000 | | | | 3,030.56 | 27,968 | | | |
| 006400.KS (SDI) | M | KRW | 430,000 | 520,000 | 92.9% | KRW | (9,933.80) | **1,751.66** | **18,577** | (43.3) | 245.5 | 23.1 |
| *OLD* | | | | | | | | 2,099.51 | 18,376 | | | |
| ASIAX | | | 1,865.18 | | | | | | | | | |

**PRICE TARGET CHANGE / ESTIMATE CHANGE IN BOLD**

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

## INVESTMENT IMPLICATIONS

Korean battery makers look well place to benefit from the growth in ESS demand in the US and the tariff wall & subsidies which have been put in place to protect them from Chinese competition. Data center demand in particular could be a material driver of growth with a 10x increase in cumulative demand through to the end of the decade. Even by 2030, the storage/capacity multiplier of 2.7x could well below the 4-8x multiplier for new data centers which are increasingly being build off grid. With LFP capacity set to expand significantly this year, Korean battery makers also look better placed to service this demand. SDI is more exposed to the ESS trend than LGES, with almost half of battery sales set to come from ESS and relatively lower EV exposure. Valuations more attractive given the sell-off driving forward EV/EBITDA multiples to 12x and long term (2030) multiples to 5x. But with questions surrounding the sustainability of AI capex and continued uncertainty on 2H26 margins we still believe there are better entry points ahead. We remain Market-Perform on LGES, LG Chem and SDI.

Model updates: we make a series of mostly minor update based on our recent demand update. For the details of the model updates, please see page 14 onwards.

## VALUATION COMPS TABLE

| Company | Price 20-Jul | Curcy | EV USD mn | Mkt cap USD mn | P/E 24A | P/E 25A | P/E 26E | P/E 27E | EV/Sales 24A | EV/Sales 25A | EV/Sales 26E | EV/Sales 27E | Sales Growth 24-25 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LGES | 319,500.0 | KRW | 69,432 | 50,510 | -109.9 | -101.9 | -2,094.1 | 45.3 | 3.9 | 4.3 | 3.5 | 2.8 | -8% |
| CATL(H) | 616.0 | HKD | 350,197 | 368,912 | 0.0 | 37.4 | 26.0 | 21.3 | 0.0 | 5.7 | 3.9 | 3.2 | 19% |
| CATL(A) | 372.5 | CNY | 240,705 | 259,421 | 36.5 | 26.0 | 18.3 | 14.8 | 4.7 | 3.9 | 2.7 | 2.2 | 19% |
| Samsung SDI | 403,500.0 | KRW | 30,136 | 22,002 | 31.9 | -60.9 | 80.5 | 23.7 | 2.5 | 3.3 | 2.8 | 2.3 | -26% |
| Panasonic | 1,607 | JPY | 23,722 | 60,363 | 20.9 | 27.1 | 38.3 | 21.5 | 0.4 | 0.4 | 0.5 | 0.5 | 7% |
| BYD | 89.9 | HKD | 130,640 | 117,066 | 21.4 | 22.7 | 20.0 | 15.7 | 1.2 | 1.1 | 1.0 | 0.9 | 15% |
| SK Innovation | 117,600 | KRW | 40,697 | 13,452 | -14.2 | -15.4 | 12.7 | 19.6 | 0.8 | 0.7 | 0.7 | 0.7 | 8% |
| | | | | | | | | | | | | | |
| Gotion High Tech | 26.0 | CNY | 15,265 | 6,968 | 59.4 | 15.9 | 25.4 | 17.3 | 3.0 | 2.3 | 1.7 | 1.3 | 29% |
| EVE | 50.6 | CNY | 20,318 | 16,222 | 27.5 | 24.7 | 15.8 | 11.9 | 3.0 | 2.1 | 1.4 | 1.1 | 41% |
| Sunwoda | 16.2 | CNY | 8,748 | 4,418 | 19.2 | 15.0 | 12.6 | 8.7 | 1.1 | 0.9 | 0.7 | 0.6 | 25% |
| CALB | 17.2 | HKD | 15,446 | 3,897 | 46.2 | 21.0 | 10.6 | 7.0 | 3.5 | 2.6 | 1.5 | 1.2 | 38% |

| Company | LTM % | Rel LTM % | LTM High | LTM Low | GPM (%) 25A | OPM (%) 25A | ND/E 24A | ND/E 25A | EV/EBITDA 24A | EV/EBITDA 25A | EV/EBITDA 26E | EV/EBITDA 27E | 25A EV/Capacity (USD M/GWh) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LGES | -1% | -55% | 437,500 | 247,000 | 18% | 6% | 40% | 48% | 28.9 | 20.5 | 18.4 | 12.3 | 196 |
| CATL(H) | 64% | 56% | 795 | 392 | 25% | 19% | -34% | -46% | 0.0 | 24.1 | 17.1 | 14.0 | 467 |
| CATL(A) | 38% | -7% | 469 | 255 | 25% | 19% | -34% | -46% | 20.4 | 16.6 | 11.7 | 9.6 | 321 |
| Samsung SDI | 144% | 8% | 820,000 | 180,100 | 8% | -13% | 42% | 34% | 16.3 | 89.4 | 18.9 | 11.7 | 479 |
| Panasonic | 182% | 102% | 4,770 | 1,411 | 30% | 5% | 11% | 15% | 4.5 | 4.1 | 5.3 | 3.7 | 395 |
| BYD | -28% | -29% | 136 | 71 | 18% | 4% | -51% | 5% | 9.3 | 9.0 | 7.1 | 6.0 | 278 |
| SK Innovation | 7% | -49% | 166,000 | 87,700 | 6% | 1% | 78% | 68% | 20.3 | 16.2 | 8.5 | 9.1 | 369 |
| | | | | | | | | | | | | | |
| Gotion High Tech | -10% | -25% | 49.8 | 24.3 | 18% | 7% | 125% | 122% | 26.3 | 16.9 | 13.8 | 11.4 | 70 |
| EVE | 16% | -26% | 94.4 | 43.4 | 17% | 8% | 37% | 49% | 22.0 | 17.0 | 12.2 | 9.7 | 157 |
| Sunwoda | -21% | -49% | 37.9 | 16.0 | 16% | 3% | 61% | 99% | 16.0 | 14.7 | 9.0 | 6.9 | 107 |
| CALB | 1% | -9% | 39.3 | 16.3 | 17% | 6% | 96% | 106% | 26.7 | 19.6 | 10.6 | 8.0 | 82 |

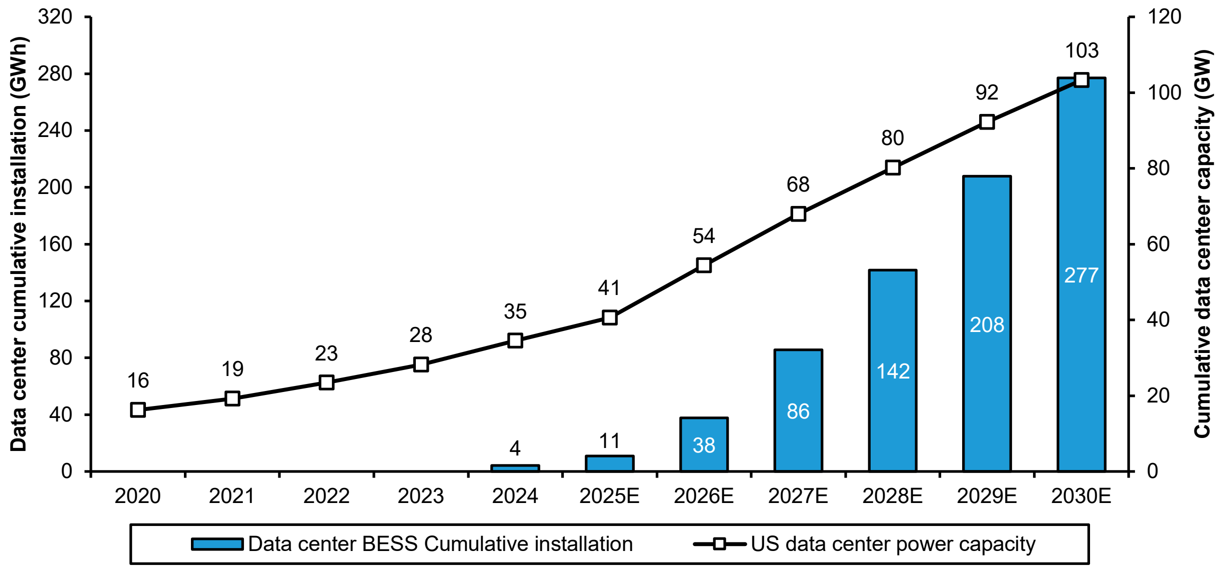
CATL, LG Energy Solution, and Samsung SDI are covered by Neil Beveridge. BYD is covered by Eunice Lee. All other stocks are not covered by Bernstein.
Source: Bloomberg (consensus), Bernstein analysis

# DETAILS

## DATA CENTER DEMAND IN US

We expect cumulative U.S. data center power capacity to reach 103 GW by 2030, driving incremental battery demand to 69 GWh in 2030 and cumulative battery installations to 277 GWh.

EXHIBIT 2: **Total capacity is expected to reach 103 GW in 2030, with cumulative battery capacity reaching 277 GWh.**




Source: BNEF, Bernstein analysis and estimates

Based on more than 20 announced U.S. data center projects, we have compiled a project-level summary table. The table shows that most projects are structured as utility-scale DC-PPAs or on-site behind-the-meter installations, with both power capacity and battery demand increasing clearly toward the end of the decade. For projects announced after 2024, the implied battery duration is generally above four hours, typically in the 4–8 hour range. This suggests that new data centers are increasingly adopting longer-duration battery solutions.

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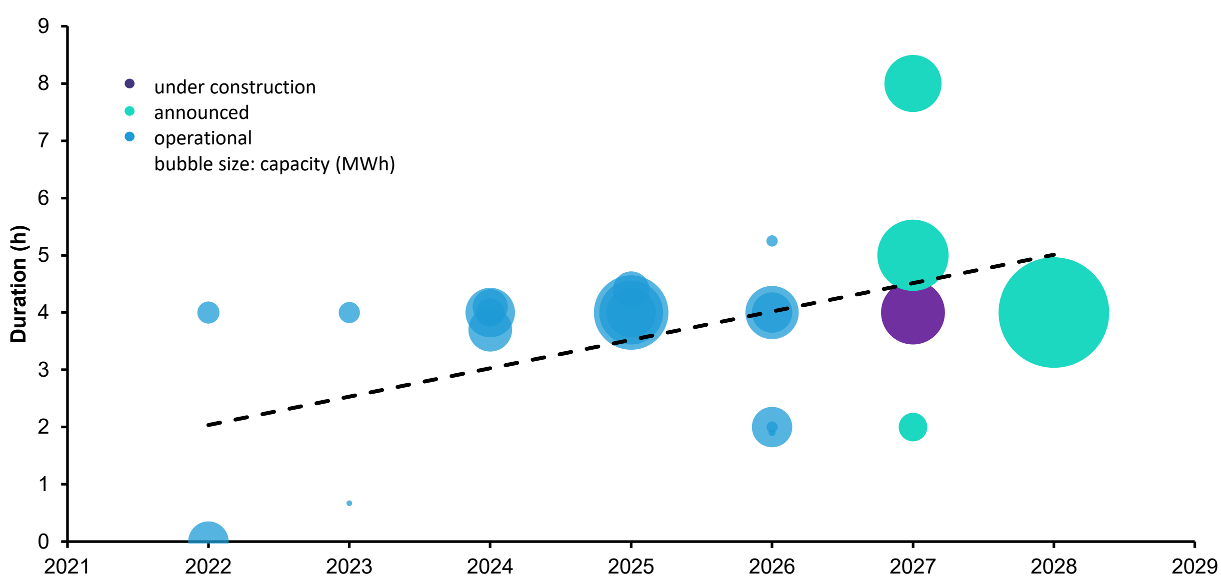
EXHIBIT 3: **Project-Level Benchmark for US Data Center BESS**

| Project Name | Company / Operator | Year (Op./Ann.) | Power (MW) | Energy (MWh) | Duration (Hours) | Status | Storage Category |
|---|---|---|---|---|---|---|---|
| Apple California Flats CA | Apple | 2022 | 60 | 240 | 4.0 | Operational | Utility-tied (PPA) |
| Switch Gigawatt 1 Nevada (4 phases) | Switch / Capital Dynamics | 2022 | 200 | 800 | 4.0 | Operational | On-site BTM (off-grid DC campus) |
| AES / Microsoft California | Microsoft / AES | 2023 | 55 | 220 | 4.0 | Operational | Utility DC-PPA |
| Microsoft Sweden (Saft) — 24 MW / 16 MWh | Microsoft / Saft (TotalEnergies) | 2023 | 24 | 16 | 0.7 | Operational | On-site BTM (UPS replacement) |
| AES Baldy Mesa + Silver Peak CA | Amazon / AES | 2024 | 100 | 400 | 4.0 | Operational | Utility DC-PPA |
| Eleven Mile Solar Center AZ (Ørsted / Fluence → Meta) | Meta / Ørsted / Fluence | 2024 | 300 | 1,200 | 4.0 | Operational | Utility DC-PPA |
| Google BESS Portfolio 2024 | Google | 2024 | 250 | 936 | 3.7 | Operational | Utility DC-PPA |
| xAI Colossus Phase 1 TN | xAI / Tesla | 2024 | 150 | 608 | 4.1 | Operational | On-site BTM |
| xAI Colossus Phase 2 TN | xAI / Tesla | 2025 | 150 | 655 | 4.4 | Operational | On-site BTM |
| AES Bellefield 1 CA | Amazon / AES | 2025 | 500 | 2,000 | 4.0 | Operational | Utility DC-PPA |
| Google Bell Solar TX (X-ELIO / Fluence) | Google / X-ELIO | 2025 | 300 | 1,200 | 4.0 | Operational | Utility DC-PPA |
| Menifee Power Bank CA (Calpine) | Multi-DC Tenants (Calpine) | 2025 | 680 | 2,720 | 4.0 | Operational | Utility DC-PPA |
| Greenflash Project Soho TX | Greenflash Infrastructure (H.I.G.) | 2026 | 400 | 800 | 2.0 | Under Construction | Utility DC-PPA |
| Iron Mountain NJE-1 NJ | Iron Mountain | 2026 | 12 | 23 | 1.9 | Under Construction | On-site BTM |
| Aligned DC + Calibrant OR | Aligned Data Centers / Calibrant Energy | 2026 | 31 | 62 | 2.0 | Under Construction | On-site BTM |
| Crusoe / Redwood Energy TX | Crusoe / Redwood Energy | 2026 | 12 | 63 | 5.3 | Operational | On-site BTM |
| Platinum Storage TX | Multi-DC Tenants (Engie) | 2026 | 348 | 1,392 | 4.0 | Operational | Utility DC-PPA |
| GridStor Oklahoma | GridStor (Goldman Sachs) | 2026 | 200 | 800 | 4.0 | Under Construction | Utility DC-PPA |
| Greenflash Rock Rose TX | Greenflash Infrastructure (H.I.G.) | 2027 | 200 | 400 | 2.0 | Announced | Utility DC-PPA |
| AES Bellefield 2 CA | Amazon / AES | 2027 | 500 | 2,000 | 4.0 | Under Construction | Utility DC-PPA |
| NIPSCO GenCo IN | Amazon / NiSource | 2027 | 400 | 1,600 | 4.0 | Under Construction | Utility DC-PPA |
| Cowboy Project WY | Meta / Enbridge | 2027 | 200 | 1,600 | 8.0 | Announced | Utility DC-PPA |
| e-Storage / Canadian Solar US | Undisclosed US Utility (DC grid) | 2027 | 500 | 2,493 | 5.0 | Announced | Utility DC-PPA |
| DTE / Oracle MI (Stargate) — 8 sites | Oracle / DTE Energy | 2028 | 1,500 | 6,000 | 4.0 | Announced | Utility DC-PPA |

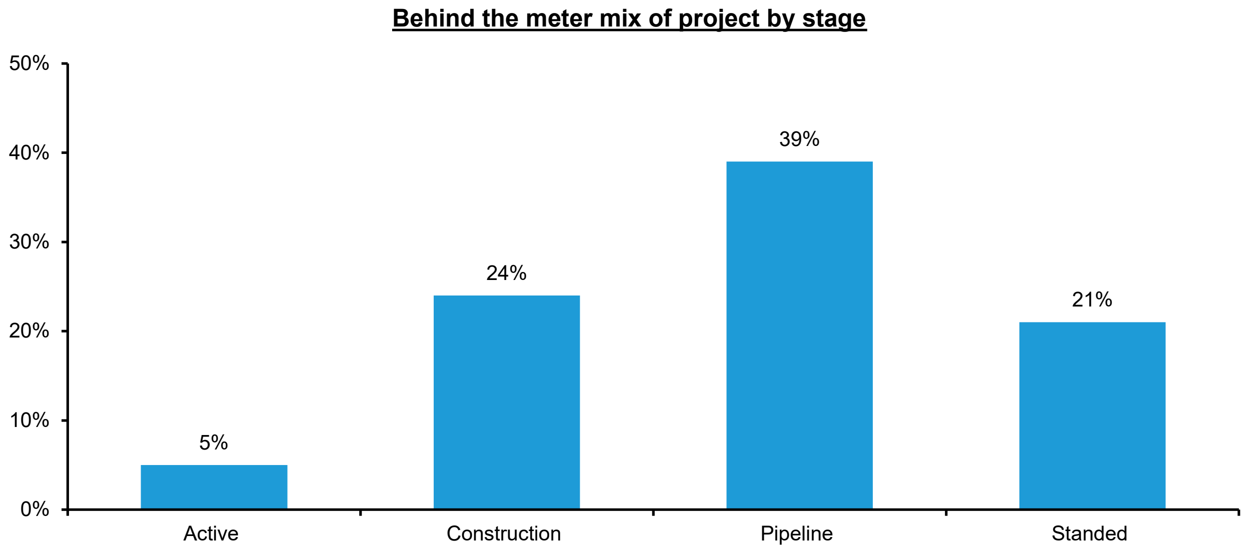
Source: Companies, Bernstein analysis

EXHIBIT 4: **Project-Level Trend of US Data Center BESS Duration and Capacity**




Source: Companies, Bernstein analysis

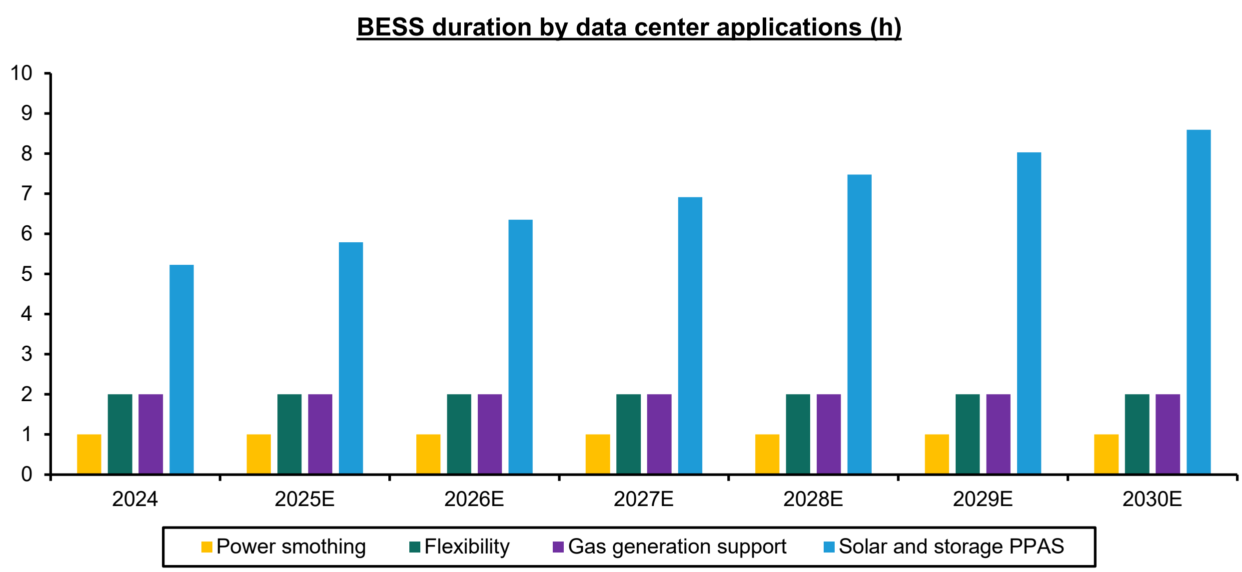
EXHIBIT 5: **Behind the meter: Only 5% of the installed base, but 24% of what's under construction, and 39% of the project pipeline**




Standed capacity = delayed+cancelled+notapproved/withdrawn capacity
Source: Aterio, Bernstein analysis

Duration requirements differ meaningfully by data center battery application. Based on BNEF data, power smoothing, flexibility, and gas generation support remain relatively stable at around 1 hour, 2 hours, and 2 hours, respectively. However, solar and storage PPAs are expected to require longer-duration BESS solutions, driving the overall duration increase over time. Based on our estimates, solar and storage PPA duration rises from approximately 5 hours in 2024 to over 8 hours by 2030, after adjusting for an attachment rate of 0.61. This suggests that solar-linked data center power procurement could become a key driver of longer-duration battery demand.

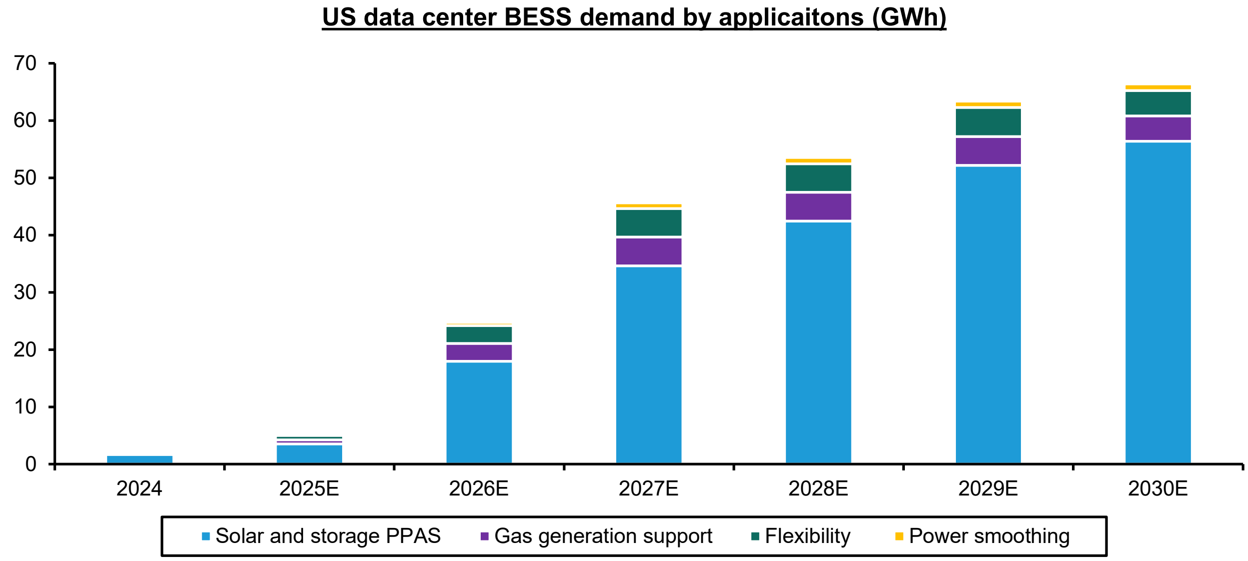
EXHIBIT 6: **BESS Duration by Data Center Application (h): Solar and storage PPAs are estimated to reach 8 hours by 2030.**




Solar duration is estimated by adjusting for the attachment rate effect, using a factor of 0.61. Source: BNEF.
Source: BNEF, Bernstein analysis and estimation

Based on our estimates, total U.S. data center BESS demand is expected to grow rapidly over the decade, reaching approximately 66 GWh by 2030. Solar and storage PPAs remain the key driver of this growth, reflecting data centers' increasing need for renewable-linked, longer-duration energy storage solutions.

EXHIBIT 7: **US data center battery installation by applications**




Source: BNEF, Bernstein analysis and estimation

Based on our data center power capacity growth assumptions, we estimate battery demand across three applications: BESS, UPS and BBU. For BESS, we assume penetration rises from 10% in 2024 to 100% by 2030, reflecting increasing adoption of energy storage for data center backup power, peak load management and renewable-linked procurement. We also assume effective BESS duration expands from 4.0 hours in 2024 to 6.0 hours in 2030, calculated as the weighted average duration across different application scenarios. For UPS and BBU, duration requirements remain relatively stable, and we assume 0.15 hours for UPS and 0.08 hours for BBU over the forecast period. On this basis, we estimate cumulative U.S. data center battery installations will reach 277 GWh by 2030, with annual additions of 69 GWh in 2030.

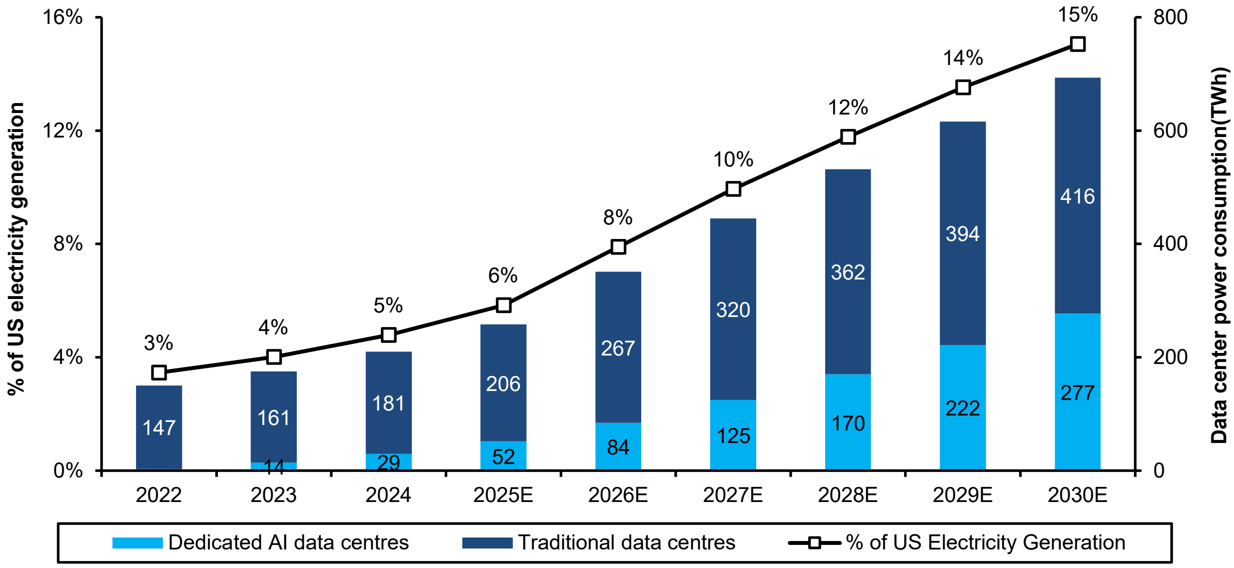
EXHIBIT 8: **US Center Battery Demand Forecast and we estimate 69GWh addition in 2030**

| | | 2020 | 2021 | 2022 | 2023 | 2024 | 2025E | 2026E | 2027E | 2028E | 2029E | 2030E |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| US data center power consumption | TWh | 100 | 118 | 139 | 173 | 218 | 258 | 351 | 445 | 532 | 616 | 694 |
| US data center power capacity | GW | 16 | 19 | 23 | 28 | 35 | 41 | 54 | 68 | 80 | 92 | 103 |
| **US data center power (incremental)** | **GW** | **2** | **3** | **4** | **5** | **6** | **6** | **14** | **14** | **12** | **12** | **11** |
| | | | | | | | | | | | | |
| BESS battery penatration | % | | | | | 10% | 20% | 40% | 70% | 85% | 95% | 100% |
| Effective capacity | GW | | | | | 1 | 1 | 6 | 10 | 10 | 11 | 11 |
| BESS duration | hrs | | | | | 4.0 | 4.2 | 4.5 | 4.8 | 5.1 | 5.5 | 6.0 |
| UPS duration | hrs | | | | | 0.15 | 0.15 | 0.15 | 0.15 | 0.15 | 0.15 | 0.15 |
| BBU duration | hrs | | | | | 0.08 | 0.08 | 0.08 | 0.08 | 0.08 | 0.08 | 0.08 |
| **Total duration** | **hrs** | | | | | **4.2** | **4.5** | **4.7** | **5.0** | **5.4** | **5.8** | **6.2** |
| | | | | | | | | | | | | |
| Data center BESS requirement | GWh | | | | | 3 | 5 | 25 | 46 | 54 | 63 | 66 |
| Data center UPS | GWh | | | | | 1 | 1 | 2 | 2 | 2 | 2 | 2 |
| Data center BBU | GWh | | | | | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| **Data center total required (incremental)** | **GWh** | | | | | **4** | **7** | **27** | **48** | **56** | **66** | **69** |
| **Data center total cumulative installation** | **GWh** | | | | | **4** | **11** | **38** | **86** | **142** | **208** | **277** |
| **Multiple** | | | | | | **0.1** | **0.3** | **0.7** | **1.3** | **1.8** | **2.3** | **2.7** |

Source: BNEF, IEA, Bernstein analysis and estimates

U.S. data centers consumed approximately 200 TWh of electricity in 2024, and we expect this to nearly triple by 2030, reaching around 15% of total U.S. power demand. Dedicated AI data centers are expected to become a key growth driver for power demand over the coming years, with electricity consumption rising from 29 TWh in 2024 to 277 TWh by 2030, accounting for around 40% of total data center power demand.

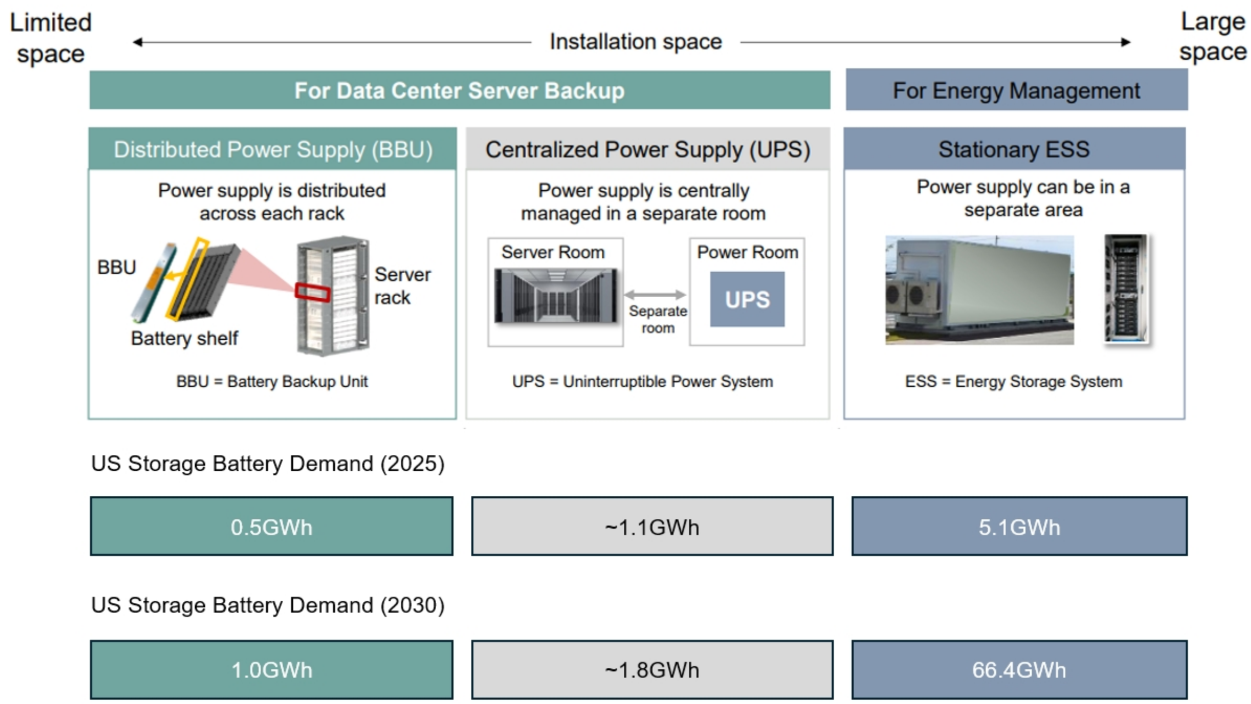
EXHIBIT 9: **US data center consumed about 200TWh in 2024 and we expected 15% of electricity consumption in 2030**




Source: BNEF, IEA, Bernstein analysis and estimates(2025+)

Among the three main data center power and backup configurations, ESS is expected to be the primary driver of battery demand growth. We estimate U.S. BBU battery demand will increase from 0.5 GWh in 2025 to 1.0 GWh in 2030, while UPS demand will rise from 1.1 GWh to 1.8 GWh. By contrast, ESS demand is expected to grow significantly from 5.1 GWh in 2025 to 66.4 GWh in 2030.

EXHIBIT 10: **Power and backup configuration of data centers. BESS used for energy management could be the key driver of battery demand**




Source: BNEF, IEA, Panasonic Energy, Synergy Research, Bernstein analysis and estimates

## US BATTERY CAPACITY AND DEMAND

Based on our updated assumptions, U.S. battery demand is now expected to grow at a CAGR of approximately 22% through 2030, supported by EV demand but increasingly driven by ESS and data center demand. However, demand growth is still expected to lag the pace of new manufacturing capacity additions. U.S. battery manufacturing capacity is projected to expand from 167 GWh at year-end 2025 to 294 GWh by year-end 2026, and could reach around 721 GWh by 2030, indicating that structural overcapacity is likely to persist through the decade.

So far in 2026, several companies have announced changes to their capacity plans. Some are shifting capacity from EV batteries to ESS, some are moving from full utilization to partial utilization, while others have pursued sale-and-leaseback arrangements to improve financial performance and reduce costs. However, most announcements provide limited detail on the scale and timing of capacity transfers or adjustments. Based on the announced changes so far, our 2030 U.S. capacity estimate has been revised down from 798 GWh to 721 GWh. We expect further announcements could help bring U.S. capacity closer to a more reasonable level.

At the same time, stronger ESS and data center demand leads us to raise our utilization rate assumptions. Our 2030 utilization rate estimate is now 15% higher than our previous forecast. We also raise our total U.S. battery demand estimate to 486 GWh in 2030, which is 74 GWh higher than our previous estimate. By segment, EV battery demand is revised lower, but this is more than offset by sharply higher demand from ESS and data centers. Driven by stronger demand growth from ESS and data centers, we estimate that ESS will account for 40% of total battery demand by 2030, with data centers contributing a further 14%.

As a result, we expect U.S. battery overcapacity to persist for most of the decade, with utilization rates remaining under pressure. This could increase the likelihood of project delays, phased commissioning schedules, or outright cancellations as producers reassess capex plans. In this context, we believe investors should focus on low-cost, well-contracted players that are better positioned to sustain cash generation in an environment where supply continues to outpace demand growth.

EXHIBIT 11: **U.S. Battery Capacity vs. Demand Outlook: Data Centers to Account for ~15% of Total Demand by 2030**

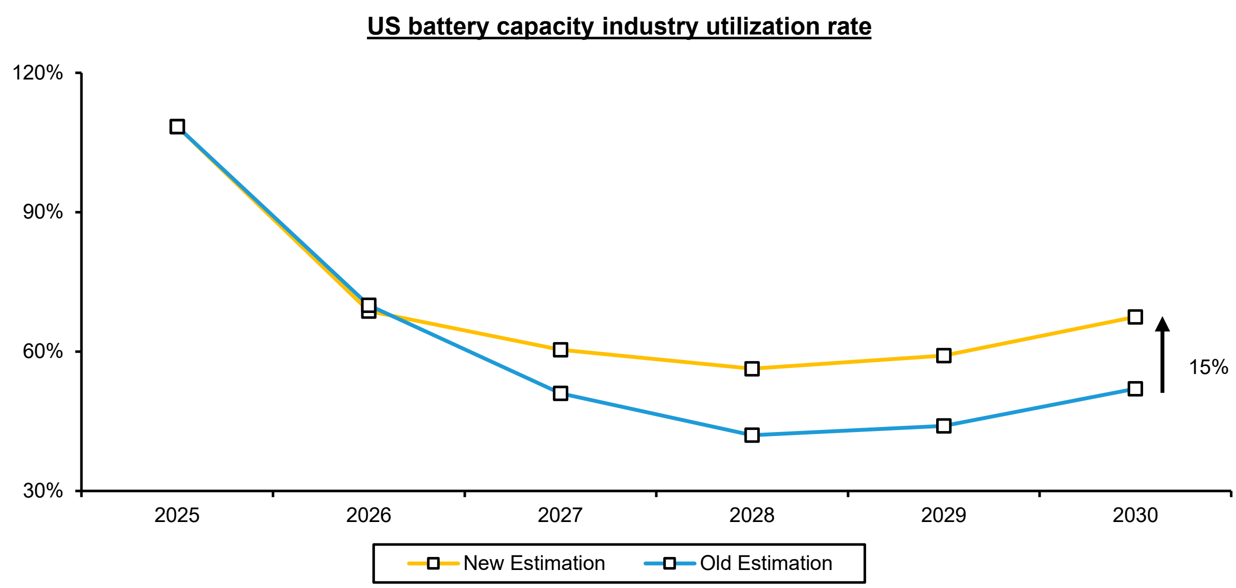
| US Market | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026E | 2027E | 2028E | 2029E | 2030E | 2025-30 CAGR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ***US Demand*** | | | | | | | | | | | | | | |
| EV | 20 | 20 | 22 | 43 | 71 | 109 | 130 | 134 | 109 | 134 | 153 | 179 | 220 | 10% |
| ESS | 1 | 1 | 3 | 11 | 12 | 21 | 34 | 42 | 66 | 108 | 166 | 175 | 197 | 36% |
| Data center | | | | | | | 4 | 7 | 27 | 48 | 56 | 66 | 69 | 59% |
| **Total Demand** | **21** | **21** | **25** | **54** | **83** | **131** | **169** | **182** | **202** | **289** | **375** | **421** | **486** | **22%** |
| Demand % YoY | | *-1%* | *19%* | *118%* | *53%* | *57%* | *29%* | *8%* | *11%* | *43%* | *29%* | *12%* | *16%* | |
| *ESS % of Demand* | *4%* | *4%* | *13%* | *20%* | *15%* | *16%* | *20%* | *23%* | *33%* | *37%* | *44%* | *42%* | *40%* | |
| ***Data center % of Demand*** | | | | | | | ***2%*** | ***4%*** | ***13%*** | ***17%*** | ***15%*** | ***16%*** | ***14%*** | |
| | | | | | | | | | | | | | | |
| LGES | 5 | 5 | 5 | 5 | 5 | 15 | 50 | 75 | 130 | 197 | 232 | 232 | 232 | 25% |
| SDI | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 15 | 42 | 64 | 64 | 73 | 37% |
| Tesla | 0 | 0 | 0 | 0 | 10 | 10 | 10 | 28 | 28 | 38 | 83 | 128 | 128 | 36% |
| SK On | 0 | 0 | 0 | 0 | 10 | 10 | 22 | 22 | 39 | 79 | 102 | 102 | 102 | 36% |
| Panasonic | 23 | 23 | 31 | 31 | 31 | 38 | 38 | 40 | 57 | 74 | 75 | 76 | 77 | 14% |
| Others | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 25 | 50 | 110 | 110 | 110 | 106% |
| **Total Available Capacity** | **31** | **31** | **39** | **39** | **59** | **76** | **123** | **167** | **294** | **479** | **665** | **711** | **721** | **34%** |
| | | | | | | | | | | | | | | |
| **Demand % Capacity** | ***68%*** | ***68%*** | ***64%*** | ***139%*** | ***141%*** | ***172%*** | ***138%*** | ***109%*** | ***69%*** | ***60%*** | ***56%*** | ***59%*** | ***67%*** | |

SDI, LGES were coverd, all others not

Source: Company data, Bernstein estimates and analysis

Driven by strong demand growth from ESS and data centers, as well as reduced capacity in the U.S., we estimate that the U.S. industry utilization rate in 2030 will be 15% higher than our previous forecast.

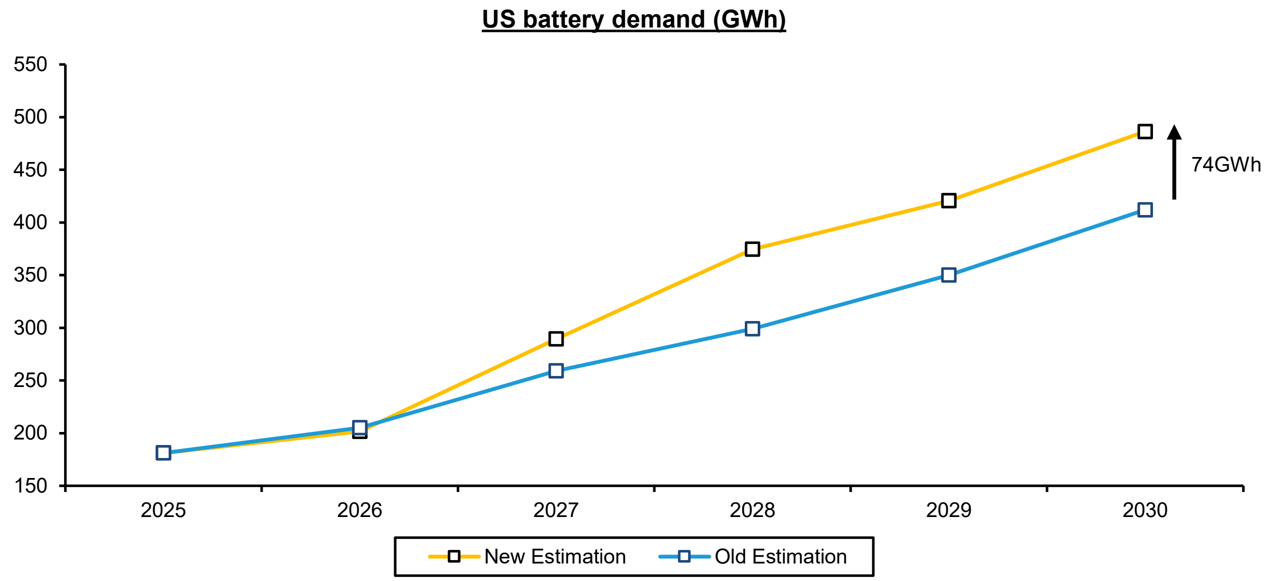
EXHIBIT 12: **We have raised our 2030 utilization rate estimate by 15% compared with our previous forecast.**




Source: Company data, Bernstein estimates and analysis

We also revise up our U.S. battery demand estimate by 74 GWh by 2030 compared with our previous forecast.

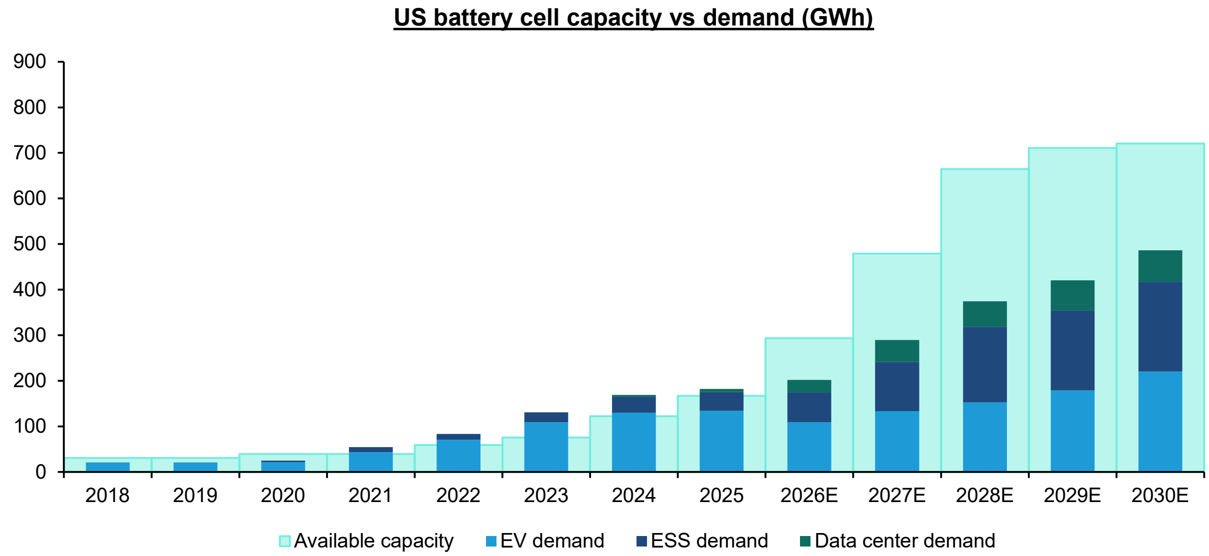
EXHIBIT 13: **We raise up our near term demand forcast and in 2030 raised 74GWh compare to last estimation**




Source: Company data, Bernstein estimates and analysis

ESS and data center demand are accelerating, while EV demand is slowing. ESS demand continues to grow rapidly, supported by rising power needs, higher renewable penetration, and policy incentives such as the ITC and AMPC. Utilities are adding storage to firm solar and wind generation, replace aging gas peakers, and improve grid reliability, particularly in markets such as California and Texas. At the same time, AI and hyperscale data centers are emerging as an important new source of battery demand, driven by backup power and peak load management needs. We estimate data centers will account for 14% of total U.S. battery demand by 2030, making this segment a key driver of incremental growth. By contrast, EV battery demand has weakened as mass-market adoption slows, subsidies are rolled back, and OEMs shift production priorities toward hybrids and ICE vehicles. Reflecting stronger ESS and data center demand, we raise our 2030 U.S. battery demand estimate by 74 GWh versus our previous forecast, with ESS and data centers expected to represent 40% and 14% of total demand, respectively, by 2030. Despite this demand uplift, supply growth remains faster than demand growth. We revise our 2030 U.S. battery capacity estimate down from 798 GWh to 721 GWh based on announced capacity adjustments, but this still exceeds our 2030 demand estimate of 486 GWh. As a result, we expect structural oversupply to persist through most of the decade. However, utilization rates could improve further if additional capacity pauses, cancellations, or delayed ramp-ups are announced.

EXHIBIT 14: **Despite strong growth in ESS and data center demand, we expect structural overcapacity in the U.S. to persist unless there are further announcements to reduce total capacity.**




Source: Company data, Bernstein estimates and analysis

Company-level capacity details are shown below. LGES remains the largest player, although we revise its capacity estimate down to 232 GWh following the pause of its Ohio battery plant. In their 1Q results conference meetings, both SDI and LGES management indicated plans to convert some EV battery production lines to ESS. However, the detailed scale and timing of these capacity adjustments have not yet been announced.

EXHIBIT 15: **US battery production capacity(year end)**

| YE US Battery Capacity (GWh) | 2024 | 2025 | 2026E | 2027E | *2030E* |
|---|---|---|---|---|---|
| LGES | 75 | 130 | 197 | 232 | 232 |
| SDI | 0 | 15 | 42 | 64 | 73 |
| Tesla | 28 | 28 | 38 | 83 | 128 |
| SK On | 22 | 39 | 79 | 102 | 102 |
| Panasonic | 40 | 57 | 74 | 75 | 78 |
| Ford | 0 | 0 | 10 | 40 | *57* |
| *Others* | 3 | 25 | 40 | 70 | *63* |
| **Total** | **167** | **294** | **479** | **665** | **732** |

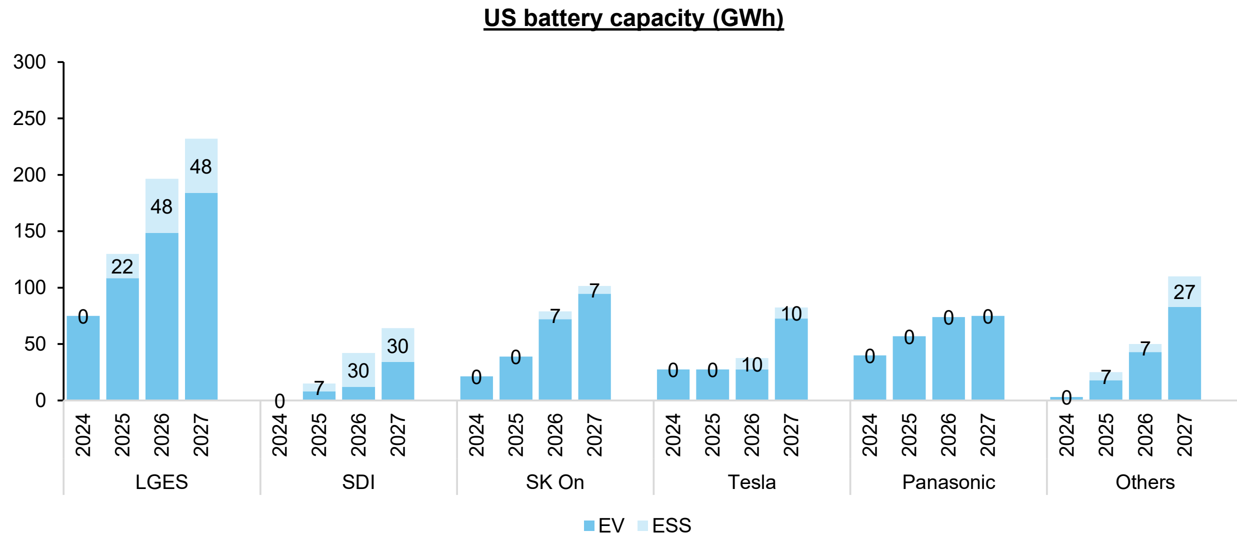
| YE US ESS Capacity (GWh) | 2024 | 2025 | 2026E | 2027E |
|---|---|---|---|---|
| LGES | 0 | 22 | 48 | 48 |
| SDI | | 7 | 30 | 30 |
| Tesla | | | 10 | 10 |
| SK On | | | 7 | 7 |
| Panasonic | | | 0 | 0 |
| Ford | | | 0 | 20 |
| *Others* | | 7 | 7 | 7 |
| **Total** | **0** | **36** | **102** | **122** |

| % ESS | 2024 | 2025E | 2026E | 2027E |
|---|---|---|---|---|
| LGES | 0% | 17% | 24% | 21% |
| SDI | | 47% | 71% | 47% |
| Tesla | 0% | 0% | 27% | 12% |
| SK On | 0% | 0% | 9% | 7% |
| Panasonic | 0% | 0% | 0% | 0% |
| Ford | | | 0% | 50% |
| *Others* | 0% | 28% | 18% | 10% |
| **Total** | **0%** | **12%** | **21%** | **18%** |

LGES, Samsung SDI are covered by Neil Beveridge. All other stocks are not covered by Bernstein
Source: Company data, Bernstein estimates and analysis

ESS cell production in the U.S. is likely to remain a bottleneck in 2026, as localized ESS supply may still lag demand from grid-scale storage projects and AI data centers. Based on our updated capacity assumptions, U.S. localized ESS battery capacity could reach around 102 GWh by year-end 2026 and increase further to around 122 GWh by 2027, supported by LGES, Samsung SDI, SK On, Tesla and other manufacturers shifting capacity toward ESS and LFP applications. LGES remains the largest U.S. battery producer, with total capacity expected to reach around 198 GWh by year-end 2026 and 232 GWh by year-end 2027. Its dedicated ESS capacity is expected to reach 48 GWh from 2026 onward. Samsung SDI appears to be making the most aggressive near-term transition to ESS, with ESS capacity expected to reach 30 GWh by 2026, representing more than half of its U.S. battery manufacturing capacity. SK On and Tesla are also expected to add ESS capacity from 2026, while other manufacturers could become a more meaningful contributor by 2027. With ESS and data center demand accelerating — and data centers expected to account for 14% of total U.S. battery demand by 2030 — localized ESS supply will likely become increasingly important. While more than 100 GWh of domestic ESS capacity by end-2026 could help meet a meaningful portion of U.S. demand and reduce reliance on China's ESS battery supply chain, ramp-up timing and execution remain key risks. Further announcements of EV-to-ESS capacity shifts could help ease supply constraints and improve the domestic ESS supply-demand balance

EXHIBIT 16: **US battery capacity by key suppliers**




SDI, LGES were coverd, All others not
Source: Company data, Bernstein estimates and analysis(2026+)

## COMPANY OUTLOOK

**Model update :**

**SDI:** We maintain our TP unchanged. However, we revise our EPS estimates, lowering FY26 EPS from 2,099 to 1,752, while slightly increasing FY27 EPS from 18,376 to 18,577. Based on the demand update we adjust the battery sales volume, but we did not change any other multiples and assumptions

**LGES:** We modestly raise our TP from KRW347,000 to KRW351,000 while maintaining our rating. We lower our FY26 EPS estimate from 1,811 to -2,364, and raise our FY27 EPS estimate from 9,453 to 10,255. Based on the demand update we adjust the battery sales volume, but we did not change any other multiples and assumptions

**LG Chem:** We modestly increase our TP from KRW298,000 to KRW312,000 while maintaining our rating. We lower our FY26 EPS estimate from 3,030.56 to 2,693, and increase our FY27 EPS estimate from 27,968 to 28,743. Based on the demand update we adjust the battery sales volume, but we did not change any other multiples and assumptions

## SAMSUNG SDI (MARKET-PERFORM, TP KRW520,000/SH)

EXHIBIT 17: **Overview of Samsung SDI's battery operation**

**Samsung SDI**

| | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 | 2025-30 CAGR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| EV | 10 | 14 | 24 | 41 | 60 | 54 | 46 | 46 | 55 | 70 | 93 | 119 | 21% |
| *%YoY* | | *40%* | *73%* | *69%* | *46%* | *-10%* | *-15%* | *0%* | *20%* | *28%* | *32%* | *28%* | |
| ESS(including data center) | 4 | 4 | 4 | 4 | 4 | 10 | 13 | 27 | 58 | 87 | 120 | 155 | 64% |
| *%YoY* | | *0%* | *-7%* | *13%* | *0%* | *138%* | *30%* | *105%* | *117%* | *50%* | *39%* | *29%* | |
| **Total battery shipments (GWh)** | **14** | **18** | **28** | **45** | **64** | **64** | **59** | **73** | **113** | **157** | **213** | **274** | **36%** |
| *%YoY* | | *29%* | *56%* | *62%* | *42%* | *0%* | *-8%* | *23%* | *55%* | *39%* | *36%* | *29%* | |
| *ESS % of Total shipment* | *29%* | *22%* | *13%* | *9%* | *7%* | *16%* | *22%* | *37%* | *51%* | *55%* | *56%* | *57%* | |
| | | | | | | | | | | | | | |
| China | 4 | 7 | 7 | 7 | 14 | 14 | 14 | 14 | 14 | 14 | 14 | 14 | 0% |
| US | 0 | 0 | 0 | 0 | 0 | 0 | 15 | 42 | 64 | 64 | 73 | 73 | 37% |
| Europe | 8 | 17 | 24 | 31 | 37 | 37 | 47 | 47 | 57 | 67 | 77 | 87 | 13% |
| Other | 6 | 6 | 6 | 12 | 12 | 12 | 38 | 38 | 38 | 38 | 38 | 38 | 0% |
| **Total battery capacity(GWh)** | **18** | **30** | **37** | **49** | **63** | **63** | **114** | **141** | **173** | **183** | **202** | **212** | **13%** |
| *%YoY* | | *67%* | *23%* | *34%* | *27%* | *0%* | *81%* | *24%* | *23%* | *6%* | *10%* | *5%* | |
| *US % of Total capacity* | *0%* | *0%* | *0%* | *0%* | *0%* | *0%* | *13%* | *30%* | *37%* | *35%* | *36%* | *34%* | |
| | | | | | | | | | | | | | |
| **Utilization** | **78%** | **60%** | **76%** | **92%** | **102%** | **102%** | **52%** | **51%** | **65%** | **86%** | **106%** | **129%** | |
| **OPM (excluding AMPC)** | **0%** | **3%** | **5%** | **7%** | **7%** | **1%** | **-16%** | **-3%** | **3%** | **6%** | **6%** | **6%** | |
| **OPM (including AMPC)** | **2%** | **3%** | **5%** | **7%** | **7%** | **2%** | **-12%** | **1%** | **10%** | **14%** | **15%** | **12%** | |

Source: Company data, SNE, Bernstein estimates and analysis(2026+estimate)

EXHIBIT 18: **We expect Samsung SDI will achieve a 4% CAGR over the period 2030-2050, while sustaining a market share of 3.4%**

| Samsung SDI M/S (in GWh) | | 2023 | 2024 | 2025 | 2035 | 2040 | 2050 |
|---|---|---|---|---|---|---|---|
| 2030-50 cagr | 3.0% | 7.2% | 5.3% | 3.4% | 2.7% | 2.7% | 2.8% |
| | **4.0%** | **7.2%** | **5.3%** | **3.4%** | **2.7%** | **3.0%** | **3.4%** |
| | 5.0% | 7.2% | 5.3% | 3.4% | 2.7% | 3.3% | 4.1% |
| | 6.0% | 7.2% | 5.3% | 3.4% | 2.7% | 3.7% | 5.0% |
| | 7.0% | 7.2% | 5.3% | 3.4% | 2.7% | 4.0% | 6.0% |

Source: Company data, SNE, Bernstein estimates (2025+) and analysis

EXHIBIT 19: **SDI target price sensitivity to revenue cagr and EBIT margin**

| Sansung SDI TP (K KRW) | | EBIT margin | | | | |
|---|---|---|---|---|---|---|
| | | 7.0% | 8.0% | 9.0% | 10.0% | 11.0% |
| 2030-50 cagr | 3.0% | 380 | 440 | 500 | 550 | 610 |
| | 4.0% | 430 | 500 | 560 | 620 | 690 |
| | 5.0% | 490 | 560 | 630 | 700 | 780 |
| | 6.0% | 560 | 640 | 720 | 800 | 880 |
| | 7.0% | 630 | 730 | 820 | 910 | 1,000 |

Source: Company data, SNE, Bernstein estimates (2025+) and analysis

## EXHIBIT 20: **Samsung SDI DCF valuation**

**Samsung SDI - Discounted Cash Flow**

| KRW bn | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 | 2025-2030 | 2031-2050E | Terminal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Revenue** | **20,124** | **22,708** | **16,592** | **13,267** | **16,072** | **22,041** | **28,333** | **35,757** | **43,352** | | | |
| Sales Growth | 48.5% | 12.8% | (26.9%) | (20.0%) | 21.1% | 37.1% | 28.5% | 26.2% | 21.2% | 26.7% | 4.0% | 3.0% |
| **EBITDA** | **3,272** | **3,365** | **2,262** | **374** | **2,064** | **3,253** | **4,414** | **5,110** | **5,714** | | | |
| **EBIT (ex-AMPC)** | **1,808** | **1,637** | **387** | **(1,673)** | **(262)** | **834** | **1,910** | **2,520** | **3,038** | | | |
| EBIT Margin | 9.0% | 7.2% | 2.3% | (12.6%) | (1.6%) | 3.8% | 6.7% | 7.0% | 7.0% | 4.0% | 8.4% | |
| **Tax** | **613** | **420** | **7** | **(495)** | **(7)** | **210** | **481** | **634** | **764** | | | |
| % of EBIT | 33.9% | 25.7% | 1.7% | 29.6% | 2.6% | 25.2% | 25.2% | 25.2% | 25.2% | 24.9% | 25.2% | |
| **NOPAT** | **1,195** | **1,217** | **381** | **(1,178)** | **(255)** | **624** | **1,430** | **1,886** | **2,274** | | | |
| **D&A** | **1,463** | **1,728** | **1,875** | **2,047** | **2,326** | **2,418** | **2,504** | **2,590** | **2,676** | | | |
| % of sales | 7.3% | 7.6% | 11.3% | 15.4% | 14.5% | 11.0% | 8.8% | 7.2% | 6.2% | 9.2% | 8.2% | |
| **Change in NWC:** | **(1,013)** | **961** | **(2,466)** | **1,585** | **-** | **-** | **-** | **-** | **-** | | | |
| % of sales | (5.0%) | 4.2% | (14.9%) | 11.9% | 0.0% | 0.0% | 0.0% | 0.0% | 0.0% | 1.0% | 0.0% | |
| **(CapEx)** | **(2,518)** | **(4,254)** | **(6,600)** | **(3,330)** | **(3,163)** | **(3,163)** | **(3,243)** | **(3,324)** | **(3,407)** | | | |
| % of sales | 12.5% | 18.7% | 39.8% | 25.1% | 19.7% | 14.4% | 11.4% | 9.3% | 7.9% | 12.4% | 8.2% | |
| **Unlevered FCF** | **(872)** | **(348)** | **(6,810)** | **(876)** | **(1,093)** | **(121)** | **691** | **1,152** | **1,543** | **1,297** | **84,402** | **98,035** |
| Our TP implied EV/EBITDA | 11.2x | 10.9x | 15.7x | 56.6x | 14.0x | 7.9x | 5.6x | 4.6x | 4.6x | | | |

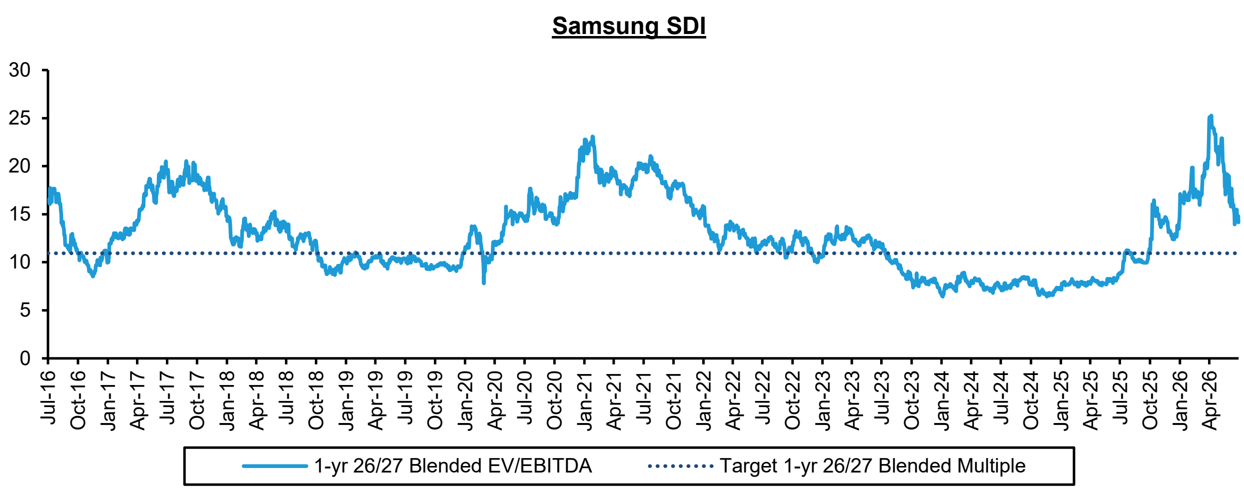
| **Valuation** | |
|---|---|
| **WACC** | **9.3%** |
| Assumed Perpetual Growth | 3.0% |
| **Enterprise Value (EV)** | **36,686** |
| - Debt | (11,236) |
| - Minority, Preferred and Others | (2,183) |
| + Cash | 1,981 |
| + Investment | 5,846 |
| + IRA credit | 3,804 |
| **Equity Value** | **34,897** |
| #shares | 67 |
| **DCF result (KRW)** | **520,000** |

| **WACC** | |
|---|---|
| Risk Free Rate | 2.5% |
| Levered Equity Beta | 2.0 |
| Market Risk Premium | 5.7% |
| Cost of Equity | 13.6% |
| After-tax Cost of Debt | 3.1% |
| Debt / (Market Cap + Debt) | 41.3% |
| **NPV** | |
| 2025-2030 | 1,341 |
| 2031-2050 | 23,677 |
| Terminal | 11,667 |
| **Total** | **36,686** |

| **Growth 2025-2030** | |
|---|---|
| Reinvestment Rate | 73% |
| ROE | 7% |
| Implied Growth Rate | 5% |
| NOPAT CAGR | 35% |
| 2030 Capex/DD&A | 1.3x |
| **NPV (%)** | |
| 2025-2030 | 4% |
| 2031-2050 | 65% |
| Terminal | 32% |
| **Total** | **100%** |

Source: Company data, Bernstein estimates and analysis

## EXHIBIT 21: **Our TP implies a 1-yr 26/27 blended EV/EBITDA of 11x for SDI**




Source: Bloomberg, Bernstein analysis

## LGES (MARKET-PERFORM, TP KRW351,000/SH)

EXHIBIT 22: **Overview of LGES's battery operation**

**LGES**

| | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 | 2025-30 CAGR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| EV | 27 | 48 | 85 | 101 | 136 | 137 | 152 | 168 | 205 | 253 | 322 | 403 | 21% |
| *%YoY* | | *80%* | *77%* | *19%* | *35%* | *1%* | *11%* | *11%* | *22%* | *23%* | *27%* | *25%* | |
| ESS(including data center) | 3 | 6 | 8 | 9 | 8 | 8 | 15 | 42 | 88 | 127 | 163 | 200 | 68% |
| *%YoY* | | *109%* | *38%* | *13%* | *-11%* | *0%* | *88%* | *180%* | *109%* | *45%* | *29%* | *23%* | |
| **Total battery shipments (GWh)** | **29** | **54** | **93** | **110** | **144** | **146** | **167** | **211** | **293** | **380** | **485** | **603** | **29%** |
| *%YoY* | | *85%* | *72%* | *18%* | *31%* | *1%* | *15%* | *26%* | *39%* | *30%* | *28%* | *24%* | |
| *ESS % of Total shipment* | *9%* | *11%* | *9%* | *8%* | *6%* | *5%* | *9%* | *20%* | *30%* | *33%* | *34%* | *33%* | |
| | | | | | | | | | | | | | |
| China | 27 | 27 | 27 | 93 | 123 | 123 | 153 | 153 | 153 | 153 | 153 | 153 | 0% |
| US | 5 | 5 | 5 | 15 | 50 | 75 | 130 | 197 | 232 | 232 | 232 | 232 | 12% |
| Europe | 40 | 70 | 70 | 70 | 90 | 100 | 115 | 115 | 115 | 115 | 115 | 115 | 0% |
| Other | 6 | 6 | 14 | 21 | 21 | 56 | 81 | 94 | 94 | 94 | 94 | 94 | 3% |
| **Total battery capacity(GWh)** | **78** | **108** | **116** | **199** | **284** | **354** | **479** | **558** | **594** | **594** | **594** | **594** | **4%** |
| *% YoY* | | *38%* | *7%* | *72%* | *43%* | *24%* | *35%* | *17%* | *6%* | *0%* | *0%* | *0%* | |
| *US % of Total capacity* | *6%* | *5%* | *4%* | *8%* | *18%* | *21%* | *27%* | *35%* | *39%* | *39%* | *39%* | *39%* | |
| | | | | | | | | | | | | | |
| **Utilization** | **38%** | **50%** | **81%** | **55%** | **51%** | **41%** | **35%** | **38%** | **49%** | **64%** | **82%** | **102%** | |
| **OPM (excluding AMPC)** | **-5%** | **-1%** | **4%** | **5%** | **4%** | **-4%** | **-1%** | **-3%** | **3%** | **6%** | **7%** | **10%** | |
| **OPM (including AMPC)** | **-5%** | **-1%** | **4%** | **5%** | **6%** | **2%** | **6%** | **4%** | **13%** | **16%** | **18%** | **20%** | |

Source: Company data, SNE, Bernstein estimates and analysis (2026+ estimates)

EXHIBIT 23: **We expect LGES will achieve a 7% CAGR over the period 2030-2050, and its market share will increase from current 10% to 13% by 2050**

| LGES M/S (in GWh) | | 2023 | 2024 | 2025 | 2035 | 2040 | 2050 |
|---|---|---|---|---|---|---|---|
| 2030-50 cagr | 1.0% | 16.2% | 12.1% | 9.6% | 6.0% | 5.1% | 4.2% |
| | 3.0% | 16.2% | 12.1% | 9.6% | 6.0% | 6.2% | 6.3% |
| | 5.0% | 16.2% | 12.1% | 9.6% | 6.0% | 7.5% | 9.2% |
| | **7.0%** | **16.2%** | **12.1%** | **9.6%** | **6.0%** | **9.0%** | **13.3%** |
| | 9.0% | 16.2% | 12.1% | 9.6% | 6.0% | 10.8% | 19.5% |

Source: Company data, SNE, Bernstein estimates (2025+) and analysis

EXHIBIT 24: **LGES target price sensitivity to revenue cagr and EBIT margin**

| LGES TP (K KRW) | | EBIT margin 7.0% | 8.0% | 9.0% | 10.0% | 11.0% |
|---|---|---|---|---|---|---|
| 2030-50 cagr | 1.0% | 107 | 128 | 150 | 172 | 194 |
| | 3.0% | 144 | 171 | 199 | 226 | 253 |
| | 5.0% | 195 | 229 | 264 | 298 | 332 |
| | 7.0% | 263 | 307 | 351 | 395 | 439 |
| | 9.0% | 354 | 411 | 468 | 525 | 582 |

Source: Company data, SNE, Bernstein estimates (2025+) and analysis

## EXHIBIT 25: **LGES DCF valuation**

**LGES - Discounted Cash Flow**

| KRW bn | 2022 | 2023 | 2024 | 2025 | 2026E | 2027E | 2028E | 2029E | 2030E | 2025-2030 | 2031-2050E | Terminal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Revenue** | **25,599** | **33,746** | **25,620** | **23,671** | **30,769** | **41,388** | **52,227** | **64,948** | **78,844** | | | |
| Sales Growth | 43.4% | 31.8% | (24.1%) | (7.6%) | 30.0% | 34.5% | 26.2% | 24.4% | 21.4% | 27.2% | 7.0% | 3.0% |
| **EBITDA** | **3,046** | **3,770** | **2,137** | **3,391** | **3,268** | **5,597** | **7,396** | **8,832** | **12,449** | | | |
| **EBIT (ex-AMPC)** | **1,214** | **1,486** | **(906)** | **(302)** | **(1,044)** | **1,231** | **2,957** | **4,327** | **7,884** | | | |
| EBIT Margin | 4.7% | 4.4% | (3.5%) | (1.3%) | (3.4%) | 3.0% | 5.7% | 6.7% | 10.0% | 5.2% | 9.0% | |
| **Tax** | **215** | **407** | **11** | **333** | **453** | **308** | **739** | **1,082** | **1,971** | | | |
| % of EBIT | 17.7% | 27.4% | (1.2%) | (110.3%) | (43.4%) | 25.0% | 25.0% | 25.0% | 25.0% | 32.5% | 25.0% | |
| **NOPAT** | **999** | **1,080** | **(917)** | **(635)** | **(1,497)** | **923** | **2,218** | **3,245** | **5,913** | | | |
| D&A | 1,832 | 2,284 | 3,043 | 3,693 | 4,312 | 4,366 | 4,439 | 4,505 | 4,565 | | | |
| % of sales | 7.2% | 6.8% | 11.9% | 15.6% | 14.0% | 10.5% | 8.5% | 6.9% | 5.8% | 8.9% | 8.0% | |
| **Change in NWC:** | **(7,949)** | **1,364** | **1,811** | **402** | **-** | **-** | **-** | **-** | **-** | | | |
| % of sales | (31.1%) | 4.0% | 7.1% | 1.7% | 0.0% | 0.0% | 0.0% | 0.0% | 0.0% | 0.1% | 0.0% | |
| **(CapEx)** | **(6,291)** | **(10,890)** | **(12,548)** | **(10,417)** | **(5,727)** | **(5,155)** | **(5,152)** | **(5,377)** | **(5,380)** | | | |
| % of sales | 24.6% | 32.3% | 49.0% | 44.0% | 18.6% | 12.5% | 9.9% | 8.3% | 6.8% | 12.7% | 8.0% | |
| **Unlevered FCF** | **(11,409)** | **(6,162)** | **(8,611)** | **(6,957)** | **(2,913)** | **134** | **1,505** | **2,373** | **5,098** | **(760)** | **233,448** | **319,661** |
| ***AMPC*** | **-** | ***646*** | ***1,566*** | ***1,526*** | ***2,379*** | ***4,144*** | ***5,595*** | ***7,553*** | ***8,081*** | | | |
| | | | | | | | | | | | | |
| Our TP implied EV/EBITDA | 31.7x | 21.7x | 26.7x | 19.2x | 17.1x | 9.9x | 7.4x | 5.9x | 4.7x | | | |

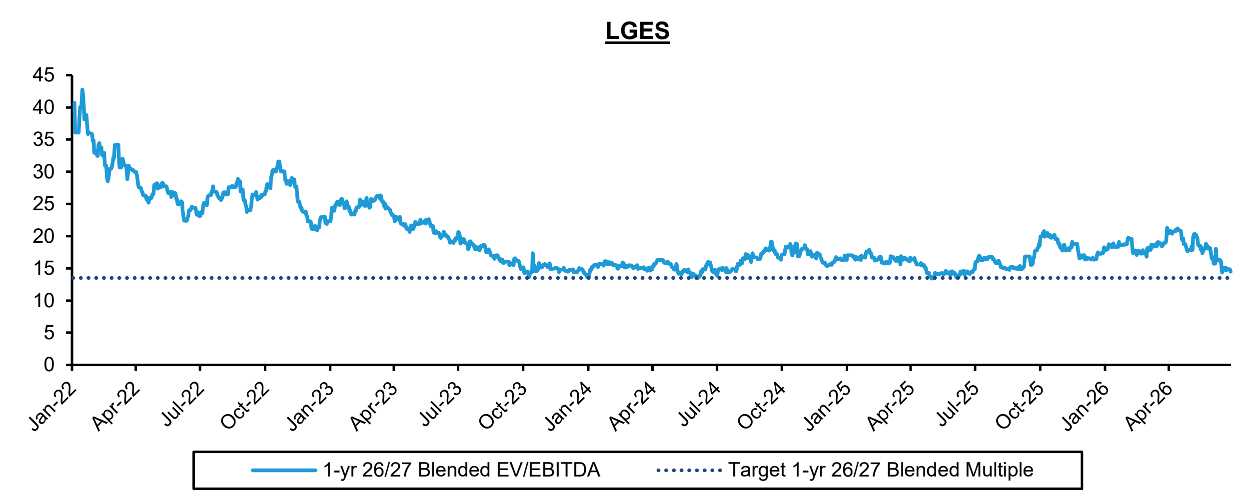
| **Valuation** | |
|---|---|
| **WACC** | **9.6%** |
| Assumed Perpetual Growth | 3.0% |
| **Enterprise Value (EV)** | **96,555** |
| - Debt | (24,682) |
| - Minority, Preferred and Others | (10,402) |
| + Cash | 2,930 |
| + IRA credit | 17,671 |
| **Equity Value** | **82,073** |
| #shares | 234 |
| **DCF result (KRW)** | $\boxed{351,000}$ |

| **WACC** | |
|---|---|
| Risk Free Rate | 2.5% |
| Levered Equity Beta | 1.5 |
| Market Risk Premium | 5.7% |
| Cost of Equity | 10.8% |
| After-tax Cost of Debt | 3.3% |
| Debt / (Market Cap + Debt) | 15.2% |
| **NPV** | |
| 2025-2030 | 3,791 |
| 2031-2050 | 57,622 |
| Terminal | 35,142 |
| **Total** | **96,555** |

| **Growth 2025-2030** | |
|---|---|
| Reinvestment Rate | 107% |
| ROE | 10% |
| Implied Growth Rate | 11% |
| NOPAT CAGR | n.a. |
| 2030 Capex/DD&A | 1.2x |
| **NPV (%)** | |
| 2025-2030 | 4% |
| 2031-2050 | 60% |
| Terminal | 36% |
| **Total** | **100%** |

Source: Company data, Bernstein estimates and analysis

## EXHIBIT 26: **Our TP implies a 1-yr 26/27 blended EV/EBITDA of 14x for LGES**




Source: Bloomberg, Bernstein analysis

### LG Chem (Market-perform, TP KRW312,000/sh)

EXHIBIT 27: **LG Chem SOTP model**

| LG Chem SOTP Segment | Avg '25-28 NP KRW bn | Multiple | Valuation Method | USD bn | KRW bn | KRW/share |
|---|---|---|---|---|---|---|
| Petrochemicals | 197 | 10.0 | P/E ('25-27) | 1.4 | 1,967 | 25,000 |
| Advanced materials | 238 | 2.0 | EV/S (2025) | 5.7 | 8,175 | 104,000 |
| Life sciences | 67 | 50.0 | P/E ('25-27) | 2.4 | 3,370 | 43,000 |
| **Energy solutions (79.38% intere:** | 1,199 | | **DCF** | **63.8** | **90,673** | **1,158,000** |
| Cash and cash eqv. | | | | 7.7 | 10,903 | 139,000 |
| Debt | | | | -23.8 | -33,812 | -432,000 |
| Less: Net Debt (Cash) | | | | 16.1 | 22,909 | 293,000 |
| **Total Value** | | | | **57.2** | **81,276** | **1,038,000** |
| **Conglomerate Discount (70%)** | | | | | | **312,000** |

Source: Company data, Bernstein estimates and analysis

EXHIBIT 28: **LGES financial exhibit**

| Income Statement (KRW bn) | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026E | 2027E | 2028E |
|---|---|---|---|---|---|---|---|---|---|
| **Revenue** | **12,570** | **17,852** | **25,599** | **33,746** | **25,620** | **23,671** | **30,769** | **41,388** | **52,227** |
| COGS | (10,568) | (13,953) | (21,308) | (28,802) | (22,214) | (19,440) | (26,058) | (33,121) | (40,391) |
| **Gross Profit** | **2,002** | **3,899** | **4,291** | **4,944** | **3,406** | **4,231** | **4,711** | **8,267** | **11,836** |
| SG&A | (2,292) | (3,130) | (3,077) | (3,456) | (4,311) | (4,533) | (3,376) | (2,892) | (3,284) |
| **Operating Profit** | **(290)** | **768** | **1,214** | **2,163** | **574** | **1,346** | **1,335** | **5,375** | **8,552** |
| Finance income/(expense) | 133 | 340 | (133) | 341 | 629 | 1,103 | 1,585 | 1,644 | 1,416 |
| Investment Income | (1) | (12) | (37) | (35) | (34) | (34) | (34) | (34) | (34) |
| Other non-operating income/(expense) | 648 | 465 | (48) | (425) | (820) | (2,000) | (1,904) | (1,200) | (1,200) |
| **Income Before Taxes** | **(701)** | **777** | **996** | **2,044** | **349** | **415** | **982** | **5,786** | **8,735** |
| Income Tax Expense (Income) | (124) | 77 | 215 | 407 | 11 | 333 | 453 | 1,157 | 1,747 |
| Profit (loss) from discontinued operations | 127 | 229 | - | - | - | - | - | - | - |
| **Net Income** | **(449)** | **930** | **781** | **1,638** | **338** | **82** | **529** | **4,629** | **6,988** |
| Minority Interest/Preferred | - | 185 | (5) | 401 | 1,357 | 1,154 | 1,082 | 2,236 | 2,365 |
| **Net Income to Common** | **(449)** | **745** | **786** | **1,237** | **(1,019)** | **(1,072)** | **(553)** | **2,393** | **4,622** |
| | | | | | | | | | |
| EBITDA | 776 | 2,220 | 3,046 | 4,447 | 3,617 | 5,039 | 5,647 | 9,741 | 12,991 |
| | | | | | | | | | |
| EPS (KRW) | (2,247) | 3,963 | 3,306 | 5,287 | (4,354) | (4,581) | (2,364) | 10,225 | 19,754 |
| BPS (KRW) | 37,827 | 57,045 | 86,751 | 104,162 | 132,338 | 125,308 | 134,140 | 153,921 | 183,784 |

| Balance Sheet (KRW bn) | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026E | 2027E | 2028E |
|---|---|---|---|---|---|---|---|---|---|
| Cash & Cash Equivalents | 1,493 | 1,283 | 5,938 | 5,069 | 3,899 | 3,779 | 2,930 | 4,647 | 8,605 |
| Accounts Receivables | 2,931 | 2,914 | 4,772 | 5,129 | 4,944 | 4,311 | 7,282 | 9,796 | 12,361 |
| Inventories | 3,043 | 3,896 | 6,996 | 5,396 | 4,552 | 4,350 | 6,117 | 7,641 | 9,518 |
| Other Current Assets | 1,618 | 1,443 | 1,443 | 1,443 | 1,443 | 1,443 | 1,443 | 1,443 | 1,443 |
| **Current Assets** | **9,086** | **9,536** | **19,149** | **17,037** | **14,838** | **13,883** | **17,772** | **23,526** | **31,927** |
| Property, plant & equipment | 8,684 | 11,051 | 15,331 | 23,655 | 38,350 | 40,795 | 43,367 | 44,126 | 44,812 |
| Intangible Assets/Goodwill | 326 | 455 | 613 | 946 | 1,534 | 1,632 | 1,735 | 1,765 | 1,792 |
| Investments | 48 | 226 | 226 | 226 | 226 | 226 | 226 | 226 | 226 |
| Other Non-current Assets | 1,798 | 2,497 | 2,981 | 3,574 | 5,360 | 10,613 | 11,642 | 11,642 | 11,642 |
| **Non-Current Assets** | **10,856** | **14,228** | **19,150** | **28,400** | **45,469** | **53,265** | **56,970** | **57,759** | **58,473** |
| **Total Assets** | **19,942** | **23,764** | **38,299** | **45,437** | **60,307** | **67,148** | **74,742** | **81,285** | **90,399** |
| | | | | | | | | | |
| Accounts Payable | 2,592 | 2,178 | 3,842 | 3,094 | 2,706 | 2,153 | 3,670 | 4,584 | 5,711 |
| Short-Term Borrowings | 1,123 | 2,203 | 2,773 | 4,185 | 6,414 | 9,975 | 11,060 | 11,060 | 11,060 |
| Other Short-Term Liabilities | 3,174 | 5,093 | 5,093 | 5,093 | 5,093 | 5,093 | 5,093 | 5,093 | 5,093 |
| **Current Liabilities** | **6,889** | **9,474** | **11,708** | **12,372** | **14,213** | **17,221** | **19,823** | **20,737** | **21,863** |
| Long-Term Borrowings | 5,078 | 4,766 | 5,336 | 6,747 | 8,977 | 12,537 | 13,622 | 13,622 | 13,622 |
| Other Long-Term Liabilities | 409 | 782 | 662 | 1,944 | 6,150 | 8,068 | 9,157 | 9,157 | 9,157 |
| **Long-term Liabilities** | **5,487** | **5,548** | **5,998** | **8,691** | **15,127** | **20,605** | **22,779** | **22,779** | **22,779** |
| **Total Liabilities** | **12,376** | **15,022** | **17,706** | **21,063** | **29,340** | **37,826** | **42,602** | **43,516** | **44,643** |
| **Shareholders' Equity** | **7,565** | **8,742** | **20,593** | **24,374** | **30,967** | **29,322** | **31,389** | **36,018** | **43,005** |
| **Total Liabilities and Equity** | **19,942** | **23,764** | **38,299** | **45,437** | **60,307** | **67,148** | **73,991** | **79,534** | **87,648** |
| *Net Debt (Cash)* | *4,708* | *5,686* | *2,171* | *5,863* | *11,492* | *18,733* | *21,752* | *20,035* | *16,077* |

| Cash Flow Statement (KRW bn) | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026E | 2027E | 2028E |
|---|---|---|---|---|---|---|---|---|---|
| Net Income | (449) | 930 | 781 | 2,092 | 338 | 82 | 529 | 4,629 | 6,988 |
| Depreciation & Amortization | 1,066 | 1,452 | 1,832 | 2,284 | 3,043 | 3,693 | 4,312 | 4,366 | 4,439 |
| Changes in Net Working Capital | 1,493 | 215 | - | - | - | - | (2,175) | (3,123) | (3,316) |
| Other | (1,407) | (1,618) | (3,193) | 68 | 1,730 | 657 | 283 | 1,000 | 1,000 |
| **Cash Flow From Operations** | **703** | **979** | **(580)** | **4,444** | **5,111** | **4,432** | **2,949** | **6,872** | **9,110** |
| Capex | (2,700) | (4,000) | (6,291) | (10,890) | (12,548) | (10,417) | (5,727) | (5,155) | (5,152) |
| Other | (947) | 1,822 | 29 | 1,170 | 482 | (465) | 628 | - | - |
| **Cash Flow From Investing** | **(3,647)** | **(2,178)** | **(6,262)** | **(9,720)** | **(12,066)** | **(10,882)** | **(5,099)** | **(5,155)** | **(5,152)** |
| Borrow/Repay | n.a. | 556 | 810 | 2,823 | 3,590 | 7,122 | 2,170 | - | - |
| Other | n.a. | 327 | 10,604 | 1,532 | 1,792 | (837) | (981) | - | - |
| **Cash Flow From Financing** | **3,164** | **883** | **11,414** | **4,355** | **5,382** | **6,285** | **1,189** | **-** | **-** |
| | | | | | | | | | |
| **Net Cash Flow** | **n.a.** | **(317)** | **4,572** | **(912)** | **(1,573)** | **(165)** | **(962)** | **1,717** | **3,958** |
| Cash at Beginning of Period | n.a. | 7,470 | 7,259 | 11,914 | 11,045 | 9,875 | 9,755 | 8,907 | 10,623 |
| FX effect | n.a. | 106 | 83 | 43 | 403 | 45 | 113 | - | - |
| **Cash at End of Period** | **1,493** | **7,259** | **11,914** | **11,045** | **9,875** | **9,755** | **8,907** | **10,623** | **14,582** |
| | | | | | | | | | |
| **Free Cash Flow** | **(1,997)** | **(3,021)** | **(6,871)** | **(6,446)** | **(7,437)** | **(5,985)** | **(2,779)** | **1,717** | **3,958** |

Source: Company data, Bernstein estimates and analysis

EXHIBIT 29: **Samsung SDI financial exhibit**

| Income Statement | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026E | 2027E | 2028E |
|---|---|---|---|---|---|---|---|---|---|
| **Revenue** | **11,295** | **13,553** | **20,124** | **22,708** | **16,592** | **13,267** | **16,072** | **22,041** | **28,333** |
| COGS | 8,914 | 10,476 | 15,903 | 18,726 | 13,498 | 11,806 | 13,007 | 16,646 | 20,560 |
| **Gross Profit** | **2,381** | **3,078** | **4,221** | **3,982** | **3,094** | **1,461** | **3,065** | **5,395** | **7,773** |
| SG&A | 1,709 | 2,010 | 2,413 | 2,349 | 2,731 | 3,183 | 2,850 | 3,155 | 3,755 |
| **Operating Profit** | **671** | **1,068** | **1,808** | **1,633** | **363** | **(1,722)** | **215** | **2,240** | **4,019** |
| Finance income/(expense) | (63) | (63) | (63) | (63) | (63) | (63) | (129) | (150) | 1 |
| Investment Income | 293 | 596 | 844 | 853 | 169 | 359 | 281 | 229 | 234 |
| Other non-operating income/(expense) | (98) | 63 | 63 | 63 | 58 | 62 | 64 | 64 | 64 |
| **Income Before Taxes** | **804** | **1,663** | **2,652** | **2,486** | **527** | **(1,364)** | **430** | **2,383** | **4,318** |
| Income Tax Expense (Income) | 173 | 413 | 613 | 420 | 7 | (495) | (7) | 465 | 842 |
| Discontinued operations | - | - | - | - | 55 | 284 | - | - | - |
| **Net Income** | **631** | **1,250** | **2,039** | **2,066** | **576** | **(585)** | **437** | **1,918** | **3,476** |
| Minority Interest/Preferred | 144 | 82 | 89 | 90 | (24) | 65 | 322 | 703 | 1,054 |
| **Net Income to Common** | **487** | **1,168** | **1,951** | **1,976** | **599** | **(650)** | **115** | **1,216** | **2,421** |
| **EBITDA** | **1,755** | **2,320** | **3,271** | **3,361** | **2,238** | **325** | **2,541** | **4,658** | **6,522** |
| EPS ex-Equity Income (KRW) | 3,819 | 10,986 | 19,900 | 19,360 | 6,727 | (13,478) | 1,905 | 15,760 | 34,123 |
| EPS - Total (KRW) | 7,439 | 17,855 | 29,810 | 30,206 | 8,917 | (9,934) | 1,752 | 18,577 | 37,006 |
| BPS (KRW) | 194,064 | 219,885 | 246,524 | 276,815 | 295,583 | 323,373 | 345,545 | 363,723 | 399,933 |
| Diluted WASO | 65.4 | 65.4 | 65.4 | 65.4 | 65.4 | 65.4 | 65.4 | 65.4 | 65.4 |

| Balance Sheet | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026E | 2027E | 2028E |
|---|---|---|---|---|---|---|---|---|---|
| Cash & Cash Equivalents | 1,669 | 2,433 | 3,097 | 2,075 | 1,885 | 1,804 | 1,981 | 1,630 | 3,760 |
| Accounts Receivables | 2,178 | 2,525 | 3,350 | 3,815 | 5,570 | 3,079 | 3,307 | 4,274 | 4,893 |
| Inventories | 1,811 | 2,487 | 3,205 | 3,297 | 2,879 | 2,936 | 2,840 | 3,365 | 3,324 |
| Other Current Assets | - | - | - | - | - | 921 | - | - | - |
| **Current Assets** | **5,657** | **7,445** | **9,652** | **9,187** | **10,334** | **8,740** | **8,128** | **9,269** | **11,977** |
| Investments | 8,576 | 9,543 | 10,145 | 11,304 | 10,187 | 11,427 | 12,372 | 12,601 | 12,835 |
| Property, plant & equipment | 6,031 | 7,544 | 8,890 | 11,862 | 17,484 | 18,935 | 20,250 | 20,995 | 21,734 |
| Intangible Assets/Goodwill | 890 | 890 | 890 | 890 | 890 | 890 | 890 | 890 | 890 |
| Other Non-current Assets | 379 | 411 | 681 | 796 | 1,702 | 2,263 | 2,715 | 2,715 | 2,715 |
| **Non-Current Assets** | **15,877** | **18,388** | **20,606** | **24,852** | **30,263** | **33,515** | **36,227** | **37,201** | **38,174** |
| **Total Assets** | **21,534** | **25,833** | **30,258** | **34,039** | **40,597** | **42,255** | **44,355** | **46,471** | **50,151** |
| Accounts Payable | 869 | 1,272 | 2,192 | 2,119 | 906 | 1,015 | 1,037 | 1,363 | 1,697 |
| Short-Term Borrowings | 2,451 | 2,527 | 2,879 | 2,894 | 6,550 | 5,498 | 5,222 | 5,157 | 5,093 |
| Other Short-Term Liabilities | 1,663 | 2,662 | 2,936 | 3,506 | 3,400 | 3,282 | 3,792 | 3,792 | 3,792 |
| **Current Liabilities** | **4,984** | **6,461** | **8,007** | **8,519** | **10,856** | **9,795** | **10,050** | **10,312** | **10,581** |
| Long-Term Borrowings | 1,508 | 2,129 | 2,341 | 2,905 | 5,190 | 6,291 | 6,014 | 5,950 | 5,885 |
| Other Long-Term Liabilities | 1,684 | 2,046 | 2,692 | 2,708 | 2,985 | 2,599 | 3,000 | 3,000 | 3,000 |
| **Long-term Liabilities** | **3,192** | **4,175** | **5,033** | **5,613** | **8,174** | **8,890** | **9,014** | **8,950** | **8,885** |
| **Total Liabilities** | **8,175** | **10,637** | **13,040** | **14,132** | **19,030** | **18,685** | **19,064** | **19,262** | **19,466** |
| **Shareholders' Equity** | **13,359** | **15,197** | **17,218** | **19,907** | **21,567** | **23,570** | **25,291** | **27,209** | **30,685** |
| **Total Liabilities and Equity** | **21,534** | **25,833** | **30,258** | **34,039** | **40,597** | **42,255** | **44,355** | **46,471** | **50,151** |

| Cash Flow Statement | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026E | 2027E | 2028E |
|---|---|---|---|---|---|---|---|---|---|
| Net Income | 631 | 1,250 | 2,039 | 2,066 | 576 | (585) | 437 | 1,918 | 3,476 |
| Depreciation & Amortization | 1,084 | 1,252 | 1,463 | 1,728 | 1,875 | 2,047 | 2,326 | 2,418 | 2,504 |
| Investment Income | (293) | (596) | (844) | (853) | (169) | (359) | (281) | (229) | (234) |
| Changes in Net Working Capital | 205 | (620) | (623) | (631) | (2,550) | 2,544 | 692 | (1,167) | (244) |
| Other | 322 | 889 | 605 | (207) | 131 | (2,636) | 85 | - | - |
| **Cash Flow From Operations** | **1,949** | **2,176** | **2,641** | **2,104** | **(138)** | **1,010** | **3,259** | **2,941** | **5,501** |
| Capex | (1,572) | (2,180) | (2,518) | (4,254) | (6,600) | (3,330) | (3,163) | (3,163) | (3,243) |
| Other | (207) | 231 | (428) | 149 | 1,680 | 747 | (28) | - | - |
| **Cash Flow From Investing** | **(1,778)** | **(1,950)** | **(2,946)** | **(4,105)** | **(4,920)** | **(2,583)** | **(3,191)** | **(3,163)** | **(3,243)** |
| **Cash Flow From Financing** | **241** | **583** | **629** | **903** | **5,544** | **1,109** | **217** | **(129)** | **(129)** |
| Cash at Beginning of Period | 1,148 | 1,559 | 2,368 | 2,692 | 1,593 | 2,080 | 1,615 | 1,793 | 1,441 |
| **Net Cash Flow** | **411** | **809** | **324** | **(1,099)** | **487** | **(465)** | **177** | **(351)** | **2,130** |
| Adjustments | 110 | 65 | 406 | 482 | (195) | 189 | 189 | 189 | 189 |
| Cash at End of Period | 1,669 | 2,433 | 3,097 | 2,075 | 1,885 | 1,804 | 1,981 | 1,630 | 3,760 |
| | | | | | | | | | |
| **Free cash flow** | **377** | **(4)** | **123** | **(2,151)** | **(6,738)** | **(2,320)** | **96** | **(222)** | **2,259** |

Source: Company data, Bernstein estimates and analysis

## EXHIBIT 30: **LG Chem financials outlook**

| **Income Statement (KRW bn)** | **2020** | **2021** | **2022** | **2023** | **2024** | **2025** | **2026E** | **2027E** | **2028E** |
|---|---|---|---|---|---|---|---|---|---|
| **Revenue** | **30,077** | **42,655** | **51,865** | **55,658** | **48,699** | **45,932** | **52,782** | **64,738** | **77,552** |
| COGS | (23,764) | (31,438) | (41,878) | (46,834) | (41,248) | (38,123) | (41,076) | (46,262) | (53,194) |
| **Gross Profit** | **6,312** | **11,219** | **9,987** | **8,824** | **7,451** | **7,809** | **11,705** | **18,477** | **24,358** |
| SG&A | (4,514) | (6,194) | (6,991) | (6,985) | (8,058) | (8,275) | (9,871) | (12,113) | (14,515) |
| **Operating Profit** | **1,798** | **5,025** | **2,996** | **2,514** | **875** | **1,182** | **1,834** | **6,364** | **9,843** |
| Finance income/(expense) | 417 | (271) | 56 | 414 | 619 | 870 | 618 | 612 | 768 |
| Investment Income | 9 | - | - | - | - | - | - | - | - |
| Other non-operating income/(expense) | 1,155 | 137 | (237) | (438) | (1,804) | (3,832) | (1,206) | (600) | (600) |
| **Income Before Taxes** | **944** | **4,891** | **2,815** | **2,490** | **(310)** | **(1,780)** | **1,246** | **6,375** | **10,011** |
| Income Tax Expense (Income) | 368 | 937 | 619 | 436 | (826) | (802) | 669 | 1,594 | 2,503 |
| Profit (loss) from discontinued operations | - | - | - | - | - | - | - | - | - |
| **Net Income** | **682** | **3,954** | **2,195** | **2,054** | **516** | **(978)** | **577** | **4,782** | **7,508** |
| Minority Interest/Preferred | 221 | 609 | 350 | 716 | 1,207 | (1) | 366 | 2,531 | 2,939 |
| **Net Income to Common** | **461** | **3,345** | **1,845** | **1,338** | **(691)** | **(977)** | **211** | **2,250** | **4,569** |
| EBITDA | 4,110 | 7,870 | 6,350 | 6,471 | 5,547 | 6,535 | 7,969 | 12,615 | 16,151 |
| EPS (KRW) | 6,666 | 47,108 | 23,670 | 17,089 | (8,826) | (13,259) | 2,693 | 28,743 | 58,366 |
| BPS (KRW) | 271,643 | 326,861 | 480,759 | 522,936 | 613,173 | 639,199 | 631,645 | 692,721 | 788,624 |

| **Balance Sheet (KRW bn)** | **2020** | **2021** | **2022** | **2023** | **2024** | **2025** | **2026E** | **2027E** | **2028E** |
|---|---|---|---|---|---|---|---|---|---|
| Cash & Cash Equivalents | 3,274 | 3,825 | 8,512 | 9,142 | 8,068 | 10,903 | 15,485 | 22,913 | 32,894 |
| Accounts Receivables | 5,298 | 6,387 | 8,957 | 8,083 | 8,166 | 6,725 | 7,982 | 9,995 | 12,143 |
| Inventories | 5,350 | 8,284 | 12,493 | 9,375 | 8,847 | 8,177 | 7,625 | 8,725 | 10,309 |
| Other Current Assets | 2,397 | 1,000 | 1,000 | 1,000 | 1,000 | 1,000 | 1,000 | 1,000 | 1,000 |
| **Current Assets** | **16,320** | **19,496** | **30,962** | **27,600** | **26,081** | **26,805** | **32,093** | **42,633** | **56,345** |
| Property, plant & equipment | 20,567 | 26,823 | 32,353 | 42,643 | 58,190 | 59,960 | 62,760 | 63,350 | 63,883 |
| Intangible Assets/Goodwill | 2,280 | 2,682 | 3,235 | 4,264 | 5,819 | 5,996 | 6,276 | 6,335 | 6,388 |
| Investments | 309 | 309 | 309 | 309 | 309 | 309 | 309 | 309 | 309 |
| Other Non-current Assets | 1,912 | 1,824 | 1,114 | 2,650 | 3,459 | 7,992 | 8,896 | 8,896 | 8,896 |
| **Non-Current Assets** | **25,069** | **31,639** | **37,012** | **49,867** | **67,777** | **74,257** | **78,241** | **78,890** | **79,476** |
| **Total Assets** | **41,389** | **51,135** | **67,974** | **77,467** | **93,858** | **101,062** | **110,334** | **121,523** | **135,822** |
| Accounts Payable | 3,459 | 3,742 | 4,756 | 4,118 | 3,682 | 3,537 | 4,106 | 4,698 | 5,551 |
| Short-Term Borrowings | 1,549 | 3,477 | 3,804 | 7,078 | 7,621 | 11,738 | 14,976 | 17,884 | 20,853 |
| Other Short-Term Liabilities | 7,617 | 7,617 | 7,617 | 7,617 | 7,617 | 7,617 | 7,617 | 7,617 | 7,617 |
| **Current Liabilities** | **12,624** | **14,836** | **16,177** | **18,813** | **18,920** | **22,892** | **26,699** | **30,199** | **34,021** |
| Long-Term Borrowings | 8,637 | 11,289 | 12,160 | 14,850 | 19,755 | 22,074 | 24,397 | 27,305 | 30,274 |
| Other Long-Term Liabilities | 1,337 | 1,800 | 2,156 | 2,866 | 7,187 | 8,990 | 9,787 | 9,787 | 9,787 |
| **Long-term Liabilities** | **9,974** | **13,089** | **14,316** | **17,716** | **26,942** | **31,064** | **34,184** | **37,092** | **40,061** |
| **Total Liabilities** | **22,598** | **27,925** | **30,493** | **36,529** | **45,862** | **53,956** | **60,884** | **67,291** | **74,082** |
| **Shareholders' Equity** | **18,790** | **23,210** | **37,481** | **40,938** | **47,996** | **47,106** | **49,450** | **54,232** | **61,740** |
| **Total Liabilities and Equity** | **41,389** | **51,135** | **67,974** | **77,467** | **93,858** | **101,062** | **110,334** | **121,523** | **135,822** |
| *Net Debt* | *6,912* | *10,941* | *7,452* | *12,786* | *19,308* | *22,909* | *23,888* | *22,276* | *18,233* |
| | - | - | - | - | - | - | - | - | - |

| **Cash Flow Statement (KRW bn)** | **2020** | **2021** | **2022** | **2023** | **2024** | **2025** | **2026E** | **2027E** | **2028E** |
|---|---|---|---|---|---|---|---|---|---|
| Net Income | 682 | 3,954 | 2,195 | 2,054 | 516 | (745) | 577 | 4,782 | 7,508 |
| Depreciation & Amortization | 2,312 | 2,845 | 3,354 | 3,957 | 4,672 | 5,353 | 6,135 | 6,251 | 6,308 |
| Changes in Net Working Capital | (875) | (3,738) | (3,646) | 1,234 | 8 | 1,966 | (136) | (2,521) | (2,879) |
| Other | 3,895 | 2,453 | (1,333) | 291 | 1,817 | 1,661 | 929 | - | - |
| **Cash Flow From Operations** | **6,014** | **5,514** | **570** | **7,536** | **7,013** | **8,235** | **7,506** | **8,512** | **10,937** |
| Capex | (5,300) | (7,100) | (9,822) | (14,301) | (14,924) | (13,310) | (7,497) | (6,900) | (6,894) |
| Other | (178) | 1,746 | 593 | 1,175 | 1,416 | 1,629 | 671 | - | - |
| **Cash Flow From Investing** | **(5,478)** | **(5,354)** | **(9,229)** | **(13,126)** | **(13,508)** | **(11,681)** | **(6,826)** | **(6,900)** | **(6,894)** |
| Borrow/Repay | 1,772 | 4,580 | 1,199 | 5,963 | 5,447 | 6,437 | 5,562 | 5,815 | 5,938 |
| Other | (826) | (4,457) | 12,133 | 195 | (556) | (194) | (1,820) | - | - |
| **Cash Flow From Financing** | **946** | **123** | **13,332** | **6,158** | **4,891** | **6,243** | **3,742** | **5,815** | **5,938** |
| **Net Cash Flow** | **1,483** | **283** | **4,673** | **568** | **(1,604)** | **2,797** | **4,421** | **7,428** | **9,981** |
| Cash at Beginning of Period | 11,558 | 16,800 | 23,004 | 31,062 | 34,440 | 31,020 | 50,773 | 76,374 | 111,730 |
| FX effect | (97) | 559 | 64 | 76 | 303 | 1,357 | 161 | - | - |
| **Cash at End of Period** | **12,944** | **17,642** | **27,741** | **31,706** | **33,139** | **35,174** | **55,355** | **83,802** | **121,711** |

Source: Company data, Bernstein estimates and analysis

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### VALUATION METHODOLOGY

#### LG Energy Solution

We value LGES using the DCF approach with a WACC of 9.6% and a terminal growth rate of 3%. Our DCF model is based on annual free cash flow forecasts until 2050, plus a terminal value estimate to capture the continuing value of the company. Our price target is KRW351,000.

#### LG Chem Ltd

We value LG Chem using a SOTP methodology to arrive at our KRW312,000 target price. We use the DCF approach with a WACC of 9.6% and a terminal growth rate of 2% to value the Energy Solutions business, which comprises ~70% of our SOTP-based target price. We value Petrochemicals business using 2025-27 P/E multiple of 10x, Advanced Materials business using a 2025 EV/Sales of 2.0x and Life Sciences business using a 2025-27 P/E of 50x.

#### Samsung SDI Co Ltd

We value Samsung SDI using the DCF approach with a WACC of 9.3% and a terminal growth rate of 3%. Our DCF model is based on annual free cash flow forecasts until 2050, plus a terminal value estimate to capture the continuing value of the company. Our price target is KRW520,000

### RISKS

#### LG Energy Solution

Key upside risks include 1) increasing market share with customer diversification which could increase sales, 2) rising margins with better costs control and raw material pass through and 3) improved battery quality which could lower the risks of recalls.

Key downside risks include 1) lower tax benefits from US IRA policy which LGES currently benefits from, 2) redistribution of tax credits with customers or partners, and 3) cost inflation in battery manufacturing.

#### LG Chem Ltd

Downside risks include: 1) increasing competition within the EV battery industry, 2) rising raw material costs which could put additional pressure on the EV value chain, 3) battery quality issues which could lead to battery recalls, and 4) a larger than expected conglomerate discount. In addition, oversupply of chemicals and weak demand could create sustained lower margins.

Key upside risks include: 1) divestment of the LGES stake; 2) faster growth in demand for EV batteries and related materials; and 3) margin expansion driven by improved cost management and effective raw material cost pass-through.

#### Samsung SDI Co Ltd

Samsung SDI's earnings growth depends on the adoption of electric vehicles and energy storage systems to boost battery revenues and profits. Any change in strategy by automakers, or lack of cost declines would reduce this upside. Other downside risks include loss of sales to OEM switching battery suppliers. A reduction of US IRA subsidies would reduce benefits to Samsung SDI's profitability which is also a downside risk.

Key upside risks comprise: 1) higher-than-anticipated demand for ESS batteries, particularly driven by robust data center requirements; 2) increased EV battery demand from clients such as BMW following the launch of new models; and 3) accelerated commercialization advancements in solid-state battery technology

**RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**

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| Market-Perform (Bernstein Brand)<br>Neutral (Autonomous Brand) | HOLD | 35.8% | 16.2% |
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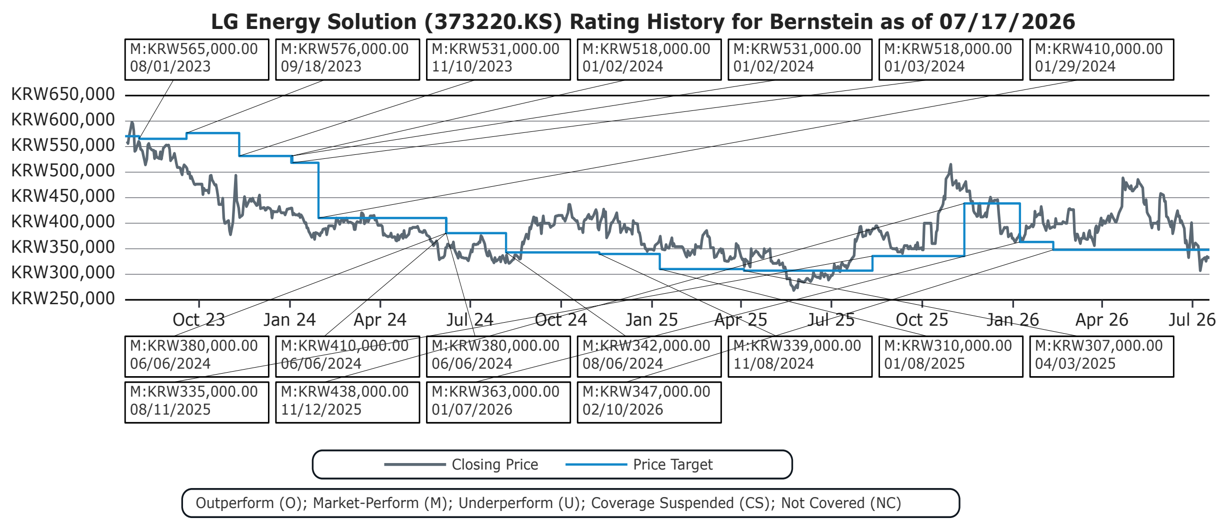


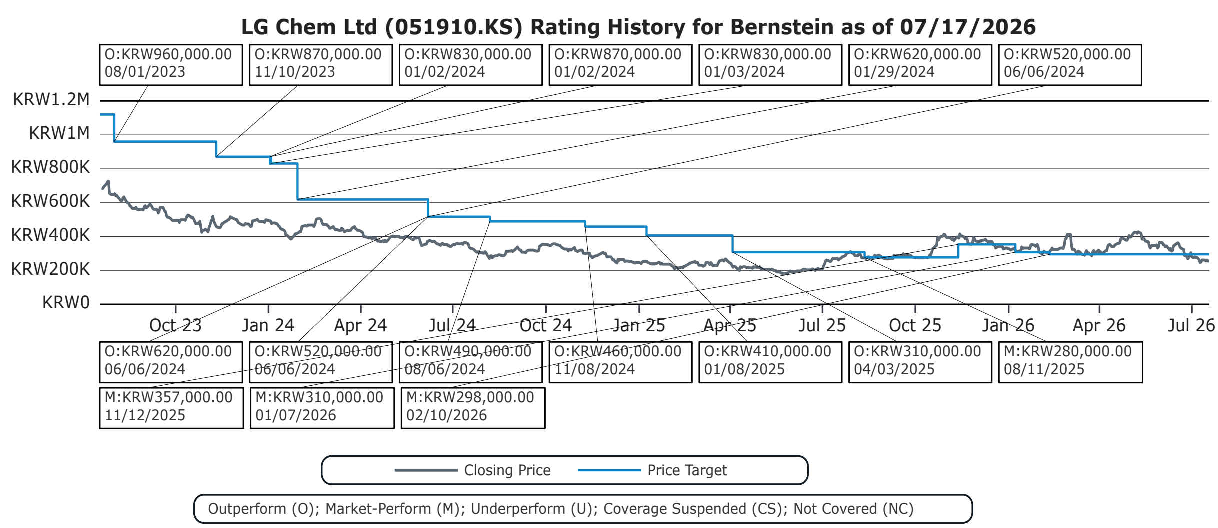

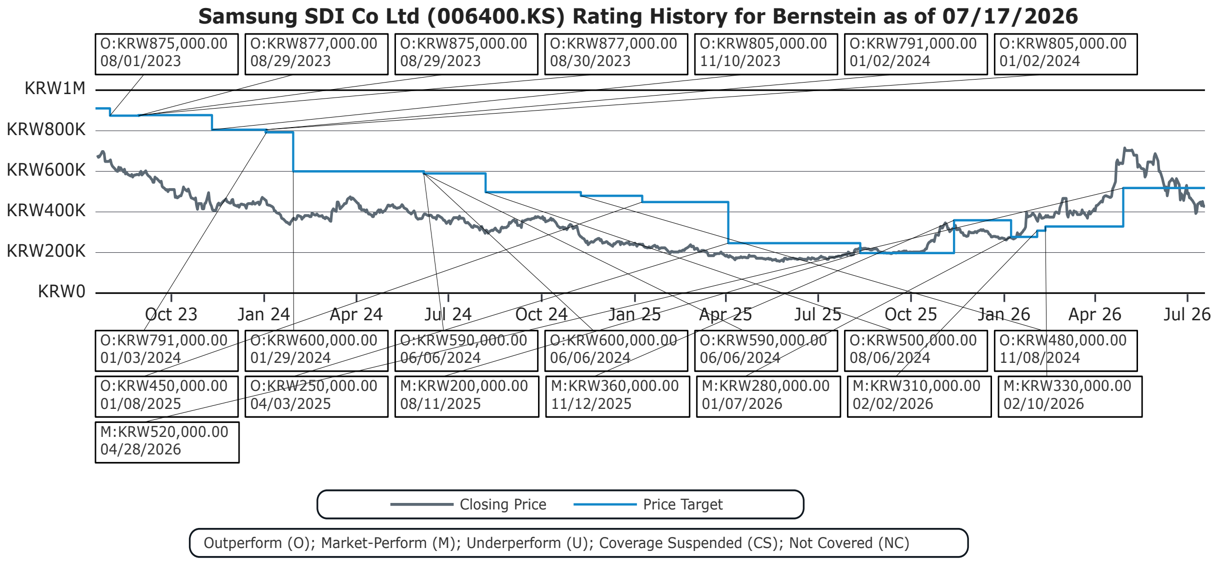


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